

Climate Change Economics  
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## ESTIMATING CLIMATE CHANGE RISK EXPOSURE: A PROBABILISTIC APPROACH

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This paper develops a methodology to quantify countries' exposure to climate change-related disasters, focusing on hydrometeorological events such as floods and storms. We introduce the Probabilistic Geospatial Risk (PGR) model, a fully Bayesian, multi-level system integrating time series, frequency, damage, and spatial sub-models. The framework captures uncertainty across the entire distribution of disaster frequency and damages, enabling robust estimation of both mean and tail risks. Using data from the EM-DAT disaster database, the World Bank, GPCC, and NOAA, we estimate disaster probabilities and damage distributions as functions of climate and development indicators. We further apply a Hilbert Space Gaussian Process (HSGP) to model fine-grained geospatial patterns of disaster occurrence within countries, complemented by state-space projections of climate and economic covariates. The model is demonstrated through a case study of the Lao People's Democratic Republic. The resulting geolocalized damage curves and return-year estimates provide a transparent, reproducible tool for assessing future exposure to climate-related risks. This probabilistic approach advances the literature by integrating uncertainty quantification, geospatial modeling, and economic impact estimation into a coherent framework suitable for policy analysis and climate adaptation planning.

*Keywords:* Climate change risk; Bayesian modeling; hydrometeorological disasters; geospatial modeling; probabilistic risk assessment; damage curves; uncertainty quantification.

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## 1. Introduction

The current paper develops a methodology to quantify countries' exposure to climate change-related disasters, and presents an example of its application for the Lao People's Democratic Republic. The methodology is implemented by one joint model called **Probabilistic Geospatial Risk (PGR) model**.

The methodology focuses on evaluating natural disasters as an important channel through which climate change impacts countries' economies. It employs a set of quantitative and probabilistic methods to estimate and predict the impact of these events.

The approach presented here advances the literature in three key ways. First, it employs a Bayesian modeling framework that enables flexible specification of probability distributions, estimation of the full distribution of disaster occurrences and damages—not just mean values—and rigorous quantification of uncertainty. Second, it incorporates a geospatial approach to connect the geographic patterns of disaster events with global climate trends. Third, it integrates several interrelated models, including time series projections of economic and climate variables, estimates of disaster frequency and severity, and spatial modeling of disaster locations. These models come together to produce probabilistic damage curves, quantifying the range of likely disaster damages in a given location, at a given time while propagating uncertainty between models.

Each step is fully Bayesian, allowing incorporation of prior knowledge held by relevant stakeholders. In addition, the structure of the model allows for scenario-based investigation. For example, it allows the incorporation of different climate forecast scenarios. The result is an end-to-end analysis pipeline that transparently generates, for a given country, region, or city, damage distributions fully characterizing tail risks faced by that region. This comprehensive perspective offers a distinct advantage over existing studies by providing a more detailed and nuanced understanding of the risk of climate-related disasters.

## 2. Literature Review

### 2.1. *CO<sub>2</sub>, climate change and disasters*

Over the past few decades, mounting scientific evidence has established a link between anthropogenic CO<sub>2</sub> emissions, global warming, and the increasing frequency and severity of natural disasters worldwide. Much of this evidence has been synthesized by the Intergovernmental Panel on Climate Change (IPCC), a scientific body formally mandated to assess climate change based on the latest scientific, technical, and socio-economic information. The IPCC's comprehensive assessments, including the Sixth Assessment Report Intergovernmental Panel on Climate Change (IPCC) (2023) and earlier publications, demonstrate how greenhouse gas (GHG) emissions, primarily from fossil fuel combustion, have driven unprecedented atmospheric temperature increases (Intergovernmental Panel on Climate Change (IPCC), 2021, 2022a,b). These temperature changes, in turn, have intensified climate-related hazards, including heatwaves, storms, floods, and droughts, all of which pose growing risks to human societies and ecosystems.

The existing literature indicates a clear causal relationship between CO<sub>2</sub> emissions and global warming, a phenomenon well-documented in the Physical Science Basis of the AR6 (Intergovernmental Panel on Climate Change (IPCC), 2021). Global surface temperatures have risen by approximately 1.1°C above pre-industrial levels, with CO<sub>2</sub> identified as the dominant driver. This warming alters the Earth's energy balance, accelerates ice melt, disrupts precipitation patterns, and amplifies the intensity of extreme weather events. The Special Report on Emissions Scenarios (Nakicenovic *et al.*, 2000) lay the groundwork for understanding how different future emission trajectories could lead to distinct climate outcomes, emphasizing the role of mitigation in curbing long-term disaster risks.

Building on this scientific foundation, the AR6 Working Group II Report (Intergovernmental Panel on Climate Change (IPCC), 2022a) highlights the growing vulnerability of human and natural systems to climate-related disasters. The report details how warming has increased the exposure of populations and infrastructure to climate hazards and stresses the compounding nature of these risks, especially in low-income regions. It also notes that socioeconomic disparities influence adaptive capacity, exacerbating disaster impacts in vulnerable communities. The way in which climate change has amplified the frequency and magnitude of natural disasters has also been documented by multiple scientific and institutional findings as indicated by Van Aalst (2006).

Finally, the AR6 Working Group III Report (Intergovernmental Panel on Climate Change (IPCC), 2022a) underlines the direct link between CO<sub>2</sub> and global warming. The report provides evidence that immediate and sustained emission reductions across all sectors are essential to staying below the 1.5°C or even 2°C warming thresholds established under the Paris Agreement. Delaying mitigation efforts increases the likelihood of irreversible climate tipping points, such as permafrost thawing or coral reef collapse, which could further exacerbate natural dis-

asters. The mitigation strategies discussed include a transition to low-carbon energy, reforestation, and carbon capture technologies.

Attributing disaster damage to climate change remains complex. Bouwer (2011) critically examine trends in economic losses and finds that much of the observed increase can be explained by rising exposure—such as more people and infrastructure in vulnerable areas—rather than a clear signal from climate change alone. This work underscores the difficulty of isolating climate-driven changes in risk from socioeconomic factors as a persistent challenge.

## **2.2. Modeling natural disasters and damage**

Numerous efforts have been developed by climate scientists to develop accurate predictions of extreme weather events independently of their economic consequences. Since this paper focuses on economic impacts, we emphasize literature addressing this aspect of the research problem.

An important focus of past research has been the development and application of Integrated Assessment Models (IAMs). However, it is important to note that IAMs often lack key components. Most notably, they do not incorporate a geospatial representation of disasters, which is central to the present research. For this reason, we adopt a different modeling approach better suited to addressing this gap.

In his Nobel Prize lecture William Nordhaus argues that climate change is the ultimate challenge for economics as an academic discipline, and asserts that “*projecting impacts is the most difficult task and has the greatest uncertainties of all the processes associated with global warming*” (Nordhaus, 2019). He also provides a clear guide through past efforts undertaken by the discipline of economics to understand and model the economic impacts of climate change.

Nordhaus (2019) emphasizes the importance of models able to integrate different systems that are deeply integrated in the real world, but that are approached with a segmented perspective by different academic disciplines. In this sense, Nordhaus recognizes the relevant role of Integrated Assessment Models (IAMs) as the powerhouse tool for economic modeling of climate change.

Botzen *et al.* (2019) develop a valuable literature review about the state of the art. One of their findings is that empirical studies on the economic impacts of natural disasters rely heavily on various data sources, with the Emergency Events Database (EM-DAT) (Centre for Research on the Epidemiology of Disasters, 2024) being the most commonly used. Compiled by the Centre for Research on the Epidemiology of Disasters (CRED), EM-DAT provides extensive data on natural disasters, although it has limitations such as variable thresholds for event inclusion and potential inflation of damage estimates. According to the authors, other databases like NatCatSERVICE and Sigma, created by reinsurance companies, are less frequently used due to their limited public availability. Recent studies have shifted towards using geophysical or meteorological variables to define natural disasters. These physical indicators provide a more objective basis for assessing the economic

impacts of disasters.

Kahn (2005) takes an innovative approach by modeling disasters frequency and impacts as a function of different economic, institutional, and development indicators. He explores the factors influencing the death toll from natural disasters across various nations. Using a comprehensive dataset on annual deaths from disasters in 57 countries from 1980 to 2002, Kahn (2005) tests several hypotheses regarding the impact of national income, geography, and institutional quality on disaster fatalities. The study reveals that while richer nations do not experience fewer natural disaster events than poorer nations, they do suffer significantly fewer deaths from these events. This suggests that economic development provides a form of implicit insurance against the adverse effects of natural disasters. Additionally, the paper finds that democracies and nations with higher quality institutions tend to have lower death tolls from natural disasters, highlighting the importance of governance and institutional frameworks in mitigating disaster impacts.

The analysis in Kahn (2005) utilizes both probit models and zero-inflated negative binomial regressions to examine the determinants of disaster fatalities. To identify the causal effect of geography on disaster fatalities, the study addresses potential confounding from national population density and institutional quality. By conditioning on democracy levels and income inequality, it isolates the specific role of geographic location. The analysis finds that geographic factors significantly influence death tolls. It also concludes that nations in Asia and the Americas experiencing higher fatalities relative to Africa. Furthermore, the paper underscores the critical role of institutional quality in insulating populations from the devastating effects of natural disasters.

López *et al.* (2015) offer a different perspective. They model the frequency of climate change related disasters as a function of country-level development indicators and global climate variables. They use a generalized linear model (GLM) to model the latent probability of a disaster with a set of country level covariates. Their approach does not consider the feedback dynamics included in IAMs models. Nevertheless, they provide a direct approach for linking climate variables, economic circumstances, and disasters.

Howard and Sterner (2017) perform a complete meta-analysis on the topic of global climate damage estimates to better understand the relationship between temperature changes and climate damages. They identify and address several biases present in previous meta-analyses, including duplicate estimates, omitted variables, measurement errors, overreliance on published estimates, dependent errors, and heteroskedasticity. They conclude that previous meta-analyses suffered from significant biases that flattened the relationship between temperature and climate damages. The authors find strong evidence that duplicate and omitted variable biases significantly impact the estimated temperature-damage relationship. Specifically, the magnitude of the bias depends on the treatment of speculative high-temperature damage estimates. When the authors replace the DICE-2013R damage function with an adjusted estimate, they find a three- to four-fold increase in the 2015 So-

cial Cost of Carbon (SCC) relative to the DICE model, depending on the treatment of productivity. When catastrophic impacts are also factored in, the SCC increases by four- to five-fold. This suggests that previous estimates of climate damages and the SCC may have been significantly underestimated due to biases in the data and methodologies used.

In conclusion, the literature in this field has grown steadily over the past two decades, but challenges still remain. The integration of granular, geospatial approaches with broader macro perspectives remains limited. Despite the advantages of adopting a probabilistic perspective to better understand disasters and their associated damages, the application of Bayesian methods remains relatively uncommon. It is along these two fronts that we believe our contribution is most relevant.

### 3. Modeling Approach

#### 3.1. *General approach*

We focus on *natural disasters* as a key channel of impact of the climate on the economy. We consider this to be a particularly relevant research problem as:

- (1) Climate researchers widely agree that disasters will likely become more frequent and more severe as the result of climate change.
- (2) Quantification of damages from discrete incidents of natural disaster is a tractable problem.
- (3) Specific actions related to disaster preparedness and prevention can be recommended to policy makers.

Our objective is to quantify countries' exposure risk to disaster events. In particular, we focus on what López *et al.* (2015) define as hydrometeorological disasters, which comprises storms, floods, and movements of wet masses. The same methodology can, however, be applied to climatological disasters, which include droughts and wildfires. Even when both disaster categories are deeply interconnected with climate dynamics, both require a separate treatment, as their causal relationships with precipitation differ significantly.

For the estimations we use a Bayesian modeling approach implemented through PyMC (Abril-Pla *et al.*, 2023), a probabilistic programming language implemented in Python. This probabilistic approach offers us three crucial advantages:

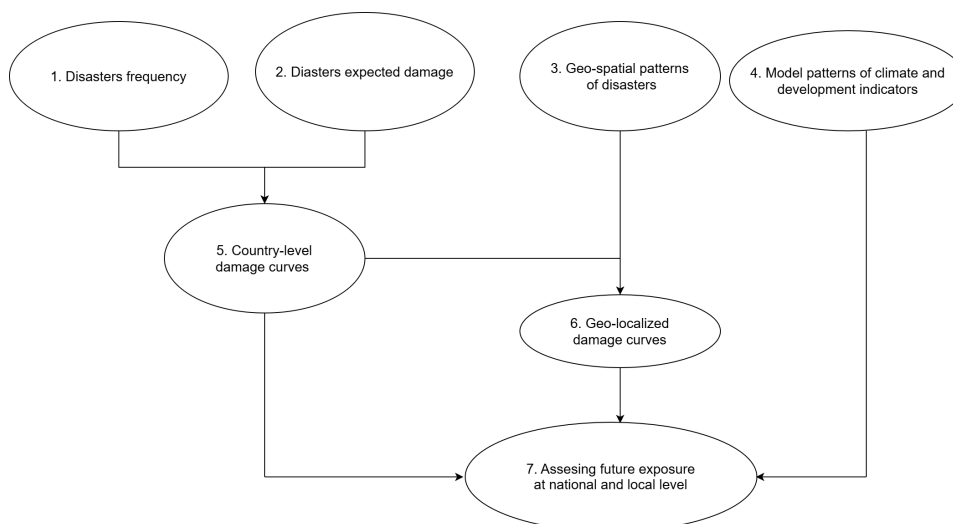
- First, the Bayesian framework facilitates the use of flexible Likelihood functions that more accurately represent the stochastic nature of disasters. Unlike traditional frequentist approaches—which often rely on asymptotic normality for inference or specific error-term assumptions (such as Gauss-Markov assumptions in OLS)—Bayesian modeling allows us to explicitly define non-normal generative processes. This is particularly relevant for disaster data, such as frequency and damage costs, which typically follow heavy-tailed or skewed distributions (e.g., Gumbel or Power Law) that are poorly captured by standard

linear frameworks.

- Second, it allows us to estimate the entire probability distributions for events frequency and expected damages instead of just the expected values. This is extremely important, as this allows us to study the tails of the probability distributions where some of the most impactful events are located. By this approach we are able to quantify country exposure to different magnitudes of disasters.
- Third, it allows us to propagate uncertainty from one piece of the model to the other. This means that rather than treating intermediate estimates (such as disaster frequencies or economic vulnerabilities) as fixed inputs, we incorporate their full uncertainty into the final output. As a result, our estimates of exposure risk more accurately reflect the real-world ambiguity and variability inherent in climate-related data.

The modeling approach consists of seven components, illustrated in Figure 1. We present here a brief description of each section, however in further sections we will develop in detail each of the steps.

Figure 1. Modeling approach



### 1. Modeling disasters frequency:

First, we use global data on disasters, climate and development indicators to model disaster frequency and predict the probability distribution of the number of disasters each country will face per year.

### 2. Modeling expected disasters damage

We use the same global data on disasters, climate, and development indicators to model the probability distribution of the expected damages for each country over

time.

### **3. Modeling geospatial patterns of disasters**

Here we take a more granular approach to model the geospatial patterns of disasters inside countries. For this we use country maps and geolocalized information of disasters, rivers, and coastlines. We combine this information with global climate time series and development indicators to compute the probability of a disaster happening for each specific point on a country map. To model the geospatial patterns, we use a Gaussian Process, a non-parametric modeling technique with great power and flexibility.

To overcome some limitations of the geospatial information available in EM-DAT data set, we build an automatized query that uses a large language model to map some of the disaster-location information from text to geospatial coordinates. We consider this to be a relevant contribution to the current methodology in the field.

In contrast to the first two models, which are estimated simultaneously using data from all countries with available observations, the geospatial component is estimated using data from a specific region only, and predictions are generated for a single target country. This approach is necessary due to the substantial computational resources required to estimate the model at a global scale. To illustrate the methodology, we present results for Laos.

### **4. Modeling patterns of climate and development indicators**

In order to assess countries' future exposure, we model the time series patterns of climate and atmospheric indicators, as well as country-level development indicators. We use Bayesian Exponential Smoothing (ETS) models with additive damped trends, estimated via Kalman filtering, to capture level and trend dynamics in the data. We use this information to build future projections of key model variables.

### **5. Country-level damage curves**

By combining the (1.) modelled disaster frequency and (2.) expected disasters damage, we can create country-level damage curves, that measure the expected damage the country will experience conditional on the expected number of disasters, and their probability distribution. These damage curve computations are expressed using the "return years" notion. Return years describe the average frequency at which a certain level of damage is expected to be met or exceeded. For example, a 100-RY event (or simply a "100-year" event) corresponds to a damage level that has a 1% chance of being surpassed in any given year, or in other words, an event of such magnitude that is expected to happen on average once every 100 years.

### **6. Geolocated damage curves**

Combining these country level damage curves (5) and the geospatial projection of the probability of a disaster (3), we build geolocalized damage curves, which measure the exposure of every point on a map to disasters.

### **7. Assessing future exposure to climate change related disasters at national and local level**

Finally, we combine all the previous outputs to build projections of event fre-

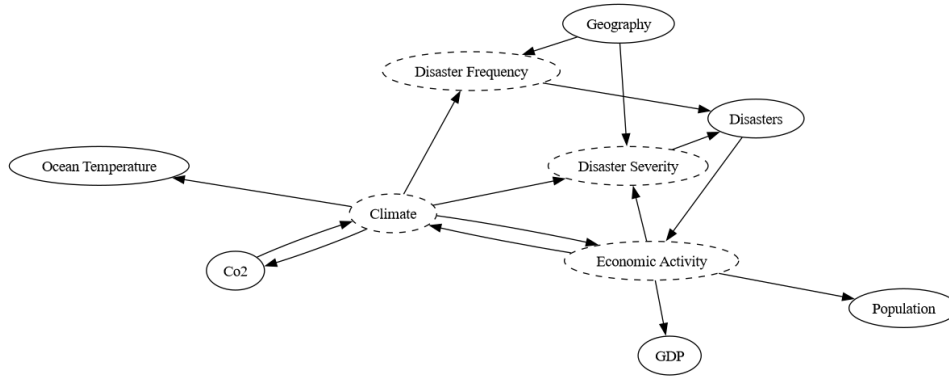


quency, damages, geospatial patterns of disaster occurrence, and country-level damage curves, as well as geolocalized damage curves.

### 3.2. Underlying causal assumptions

To reason about our causal assumptions, we employ a causal model à la Pearl (2009). Figure 2 shows the causal assumptions underlying our approach. “Economic Activity” and “Climate”, as complex, abstract systems, are unobserved processes. Instead, we observe proxies for these: GDP or population growth (in the case of the economy), or climatic indicators like atmospheric CO<sub>2</sub> or ocean temperatures (in the case of the climate). The climate, together with the location of a specific place, determine the frequency and intensity of disasters. Economic activity also contributes to disaster severity. This occurs when resources are being deployed towards adaptation and resilience (e.g. with dykes, levees, and floodwalls), or when overexploitation creates new risks (such as mudslides resulting from overdevelopment). More economic activity and higher GDP also imply more potential for damages, either from destruction of expensive infrastructure or from interruptions to economic activity. Of course, we don’t actually observe the true frequency ( $f_P(p)$ ) or severity ( $f_D(d)$ ) variables; these we are left to infer by observing the disasters that do occur.

Figure 2. Causal Graph Between Economic Activity, Climate, and Disasters. Dotted circles represent unobserved variables. Arrows connect concepts causally, so that  $A \rightarrow B$  implies that  $A$  causes  $B$ .



The causal line between economic activity and disaster severity has a more quotidian interpretation, related to our definition of “severity”. That is, we adopt an economic definition, wherein the severity of a disaster is measured by the cost of the damage it inflicts. As such, there must first be things to damage as a necessary precondition. The more things that exist, the more things that can be destroyed, and the greater potential for disaster impact.

Importantly, Figure 2 is silent on the specific functional relationships between variables. Each arrow on this diagram represents an entire discipline of study as shown in the literature review. Our objective is not to model these causal linkages individually, but to model them jointly.

## 4. Methodology

### 4.1. *Uncertainty quantification*

Uncertainty quantification lies at the heart of our modeling approach. Disaster damage data typically exhibits positive skew and long right tails, where a small number of extreme events drive the majority of observed impacts. Consequently, traditional mean-based metrics fail to capture the reality of catastrophic risk. Instead, it is the tails of the distribution that demand attention. To address this, we employ generative probabilistic modeling (Gelman *et al.*, 1995) as a framework for causal inference. This approach allows us to move beyond passive observation and perform counterfactual interventions using Judea Pearl’s do-operator (Pearl, 2009). By simulating the full distribution of potential outcomes under the intervention  $P(\text{Damage} \mid \text{do}(\text{Policy}))$ , we can conduct rigorous scenario analysis that accounts for functional relationships, prior research, and the inherent uncertainty in the data.

We implement all models using PyMC (Abril-Pla *et al.*, 2023), a Bayesian generative modeling framework. In this approach, we specify probability distributions over model parameters (priors), update them with observed data via Bayes’ theorem to obtain posterior distributions, and approximate these posteriors using Markov Chain Monte Carlo (MCMC) sampling—specifically the No-U Turn Sampler (NUTS) (Hoffman *et al.*, 2014). Our central object of interest is the *posterior predictive distribution*, which marginalizes over parameter uncertainty to produce a full probability distribution over possible outcomes. This allows us to quantify not just expected damages but also tail risks and extreme event probabilities, which are critical for building damage curves and their projections.

### 4.2. *Modeling disaster frequency*

As a first step, we model disaster frequency at country level. We use disaster data from the EM-DAT data set (Centre for Research on the Epidemiology of Disasters, 2024). We model the number of disasters a country experiences per year as a function of climate variables and country-level development indicators. We focus on one type of disasters: hydrometeorological (storms, floods, and movements of wet masses).

To model the number of disasters we use a Zero Inflated Binomial probability distribution, which allows us to better capture the fact that the generative process of disasters has a big mass concentration around zero, as multiple years do not report disasters for several countries. Consequently, we can say that the number of disasters occurring in country  $i$  on year  $t$  is distributed according to the Zero Inflated Binomial probability distribution:

$$Y_{i,t} \sim \text{ZeroInflatedBinomial}(\Psi, \mu, \alpha) \quad (1)$$

The distribution is defined by three parameters: the inflation parameter  $\Psi$  (probability of excess zeros), the mean  $\mu$  (expected count conditional on the Negative Binomial process), and the dispersion  $\alpha$ . We model  $\Psi$  using a Beta prior and  $\alpha$  using a Gamma prior. The expected count  $\mu = \mathbb{E}[Y_{i,t} | X, \theta]$  is the core of the model, linked to observed covariates through a GLM structure. We use a softplus link function to ensure positive rates, computing  $\mu_{i,t} = \text{softplus}(Z_{i,t})$  for every country  $i$  in every year  $t$ . The latent rate  $Z$  is defined as the sum of three components:

$$Z = \text{country\_fixed\_effect}_i + \text{spatial\_random\_effect}_i + \text{covariates\_effect}_{i,t} \quad (2)$$

These components have the following meanings:

- `country_fixed_effect` is a model intercept with a country hierarchy. This means that it is modeled using a Bayesian hierarchical framework, which serves as an intermediary between the extremes of no-pooling (independent OLS per country) and complete pooling (a single global intercept). This structure assumes that country-level intercepts are drawn from a common global distribution, allowing for partial pooling of information across units. For countries with sparse data or extreme outliers, the hierarchical prior induces shrinkage of the country-specific estimates toward the global mean. This “information sharing” significantly improves the reliability of estimates for small-sample countries, effectively reducing the noise and over-fitting.
- `spatial_random_effecti` accounts for spatial autocorrelation using a Besag-York-Mollié (BYM) specification (Besag *et al.*, 1991). The BYM decomposes the spatial effect into an unstructured term  $\tilde{\theta}_i$  (capturing country-specific deviations) and a structured ICAR term  $\phi_i$  (smoothing neighboring countries via an adjacency matrix  $W$ ), combined as:

$$\text{spatial\_random\_effect}_i = \sqrt{1 - \rho} \cdot \tilde{\theta}_i + \sqrt{\frac{\rho}{s}} \phi_i \quad (3)$$

where  $\rho$  is a mixing coefficient and  $s$  a scaling factor for the ICAR variance. For the `covariates_effecti,t` we introduce the time dimension  $t$ , and we use the following covariates:

- `ln_population_density_standardizedi,t`: log of population density for country  $i$ , period  $t$ .
- `ln_gdp_pc_standardizedi,t`: log of GDP per capita for country  $i$ , period  $t$ .
- `population_standardizedi,t`: Population in millions for country  $i$ , period  $t$  squared.

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- `precip.deviation.standardizedi,t`: Deviation of precipitation on country *i* on period *t* from its trend.
- `co2.standardizedt`: Worldwide atmospheric CO2 concentration in parts per million at period *t*.

To control for unobserved heterogeneity on the panel structure, we apply the Mundlak adjustment by introducing group-level (country-level) random slopes for each covariate, which allows the model to account for correlation between covariates and latent group-specific effects. This way, the covariates functional form has the structure:

$$\text{covariates\_effect}_{i,t} = \sum_{j=0}^n X_{i,t,j} \cdot \beta_j + \sum_{j=0}^n \lambda_{i,j} \cdot \beta_j^{\text{latent}} \quad (4)$$

Where:

- *i* corresponds to country index, *t* year index, and *j* covariate index.
- $\lambda_{i,j}$  is a country-specific deviation (random slope) for covariate.

#### 4.3. Modeling disaster damages

For disaster damage, we use the logarithm of disaster damage. We model this variable using a Hurdle LogNormal probability distribution. The Hurdle LogNormal distribution is a type of hurdle model used to handle zero-inflated continuous data. Hurdle models are particularly useful when dealing with datasets that have an excess of zero values, in addition to continuous positive values. The Hurdle LogNormal model specifically combines a binary process that models the zeros and a truncated LogNormal process that models the positive continuous values.

The overall likelihood for the Hurdle LogNormal model is the product of the likelihoods from the binary component and the truncated LogNormal component:

- $p$  if  $y_i = 0$
- $(1 - p) \cdot \text{LogNormal}(y_i | \mu, \sigma)$  if  $y_i \neq 0$

Here, the LogNormal component is completely defined by the parameters  $\mu$  and  $\sigma$ . To model sigma, we use a Half Normal distribution, which ensures strictly positive terms. While for  $\mu$ , we use exactly the same linear structure we used for frequency  $\mu$  on equation 10.

#### 4.4. Modeling geospatial patterns of disasters

As mentioned previously, the geospatial component is estimated using data from a specific region only, and predictions are generated only for a single target country at a time. This approach is necessary due to the substantial computational resources required to estimate the model at a global scale. To illustrate the methodology, we present results for Laos.

In the case of the probability of the event model, the objective is to model the probability of one event happening at any point of a country's grid. To achieve this, we create a set of synthetic non-disaster points, and we train the model to identify the probability of a point being a real disaster or a synthetic non-disaster point.

We estimate two different model versions. The first uses only the coordinates (latitude and longitude) of the grid and disaster points to capture geographical patterns of disaster occurrence. It captures these patterns by applying a Hilbert Space Gaussian Process (HSGP).

The second one incorporates, on top of the HSGP model, the distance to rivers and coastlines variables, plus a set of country-level covariates and climate-related time series.

In both cases, we estimate the probability of occurrence of an event by using a logit regression estimated in a Bayesian framework with PyMC (Abril-Pla *et al.*, 2023). Therefore, we model the probability of an event happening at location  $g$  in country  $i$  at time  $t$ . This way, the general specification corresponds to the general form:

$$P(Y = 1 \mid X)_{i,g,t} = \sigma(\text{country\_fix\_effect}_i + \text{HSGP}_{i,g} + X_{i,g,t} \cdot \beta_{i,g,t}) \quad (5)$$

Where:

- $\sigma(Z)$  corresponds to the logistic function  $\frac{1}{1+e^{-z}}$ .
- $\text{country\_fix\_effect}_i$  represents the country fix effect for country  $i$ .
- $\text{HSGP}_{i,g}$  is the Hilbert Space Gaussian Process component that will be explained next.
- $X_{i,t} \cdot \beta_{i,t}$  corresponds to a set of covariates and their parameters, where we include three types of covariates: geospatial, climate time series, and country-level macro variables.

### Hilbert Space Gaussian Process

We approximate the spatial component using a Hilbert Space Gaussian Process (HSGP) (Solin and Särkkä, 2020), which replaces the standard  $O(N^3)$  Gaussian Process computation with a finite basis function expansion. The HSGP projects the GP onto a set of pre-computed basis functions derived from the spectral decomposition of the covariance kernel, reducing computational cost to  $O(Nm)$  where  $m \ll N$  is the number of basis functions. We use a Matérn-5/2 kernel, which produces smooth but flexible spatial surfaces, with learned length-scale and amplitude parameters. The HSGP takes latitude and longitude as inputs and learns the spatial pattern of disaster occurrence directly from the data, capturing regional variations without requiring explicit geographic covariates. This enables scalable inference over large spatial grids while preserving the uncertainty quantification properties of the full GP.

### Full model specification

In this second step, we estimate the full model specification:

$$P(Y = 1 | X)_{i,g,t} = \sigma(\text{country\_fix\_effect}_i + \text{HSGP}_{i,g} + X_{i,g,t} \cdot \beta_{i,g,t}) \quad (6)$$

Here, in addition to the geospatial pattern detected by the HSGP, we obtain for every point in the grid of the region analyzed and for every disaster point the variables corresponding to 3 sets of covariates: climate change series, country-level specific variables, and distances to rivers and coastlines. All the covariates we introduce here are standardized to simplify the scale manipulation and the prior setting in the estimation. The standardization is done based on the whole model data set's mean and standard deviation. All the priors of the covariates parameters are defined as normal distributions with mean zero and standard deviation one.

The full set of covariates is:

Climate change series:

- `co2_standardizedt`: Worldwide atmospheric CO2 concentration in parts per million at period  $t$ .
- `dev_ocean_temp_standardizedt`: Deviation of the mean world's ocean temperature from its trend on period  $t$ .
- `precip_deviation_standardizedi,t`: Deviation of precipitation on country  $i$  on period  $t$  from its trend.

Country-level specific variables:

- `ln_population_density_standardizedi,t`: log of population density for country  $i$ , period  $t$ .
- `ln_gdp_pc_standardizedi,t`: log of GDP per capita for country  $i$ , period  $t$ .
- `square_ln_gdp_p_standardizedi,t`: log of GDP per capita for country  $i$ , period  $t$  squared.
- `population_standardizedi,t`: Population in millions for country  $i$ , period  $t$  squared.

Distances to rivers and coastlines:

- For each point of the map, we compute the distance with the closest river, and we introduce its log as feature: `log_distance_to_river_standardizedg`.
- We also compute the distance to the closest coastline, and we include its log as feature: `log_distance_to_coastline_standardizedg`.

#### 4.5. *Modeling patterns of climate and development indicators*

With the objective of being able to make predictions, we use a State Space modeling framework to predict trends in key inputs of the model. In particular, we use the implementation of State Space modeling via PyMC (Grabowski, 2023). In particular we project for a specific country of analysis: real GDP, population, deviation from trend precipitation. Additionally we project the global variables: atmospheric CO<sub>2</sub> accumulation, and ocean temperature deviations from trend.

For this, we employ a Bayesian dynamic linear model based on the PyMC state-space module, using a Bayesian Exponential Time Smoothing (ETS) (Hyndman *et al.*, 2008) approach. This approach enables probabilistic inference and forecasting under a fully Bayesian framework. We model the multivariate time series using an additive trend exponential smoothing state-space model with the following configuration: local level (A), damped local trend (Ad), and no seasonal component (N). The model specification includes measurement error, a dense innovation covariance structure for the state shocks, and a stationary initialization with dampening. This setup provides flexibility to accommodate a wide range of time-varying behavior, including long-run stochastic trends, persistent shocks, and changing volatilities across the multiple indicators.

Formally, let  $y_t \in \mathbb{R}^n$  denote the observed vector of the  $n$  variables at time  $t$ , which evolves according to a linear Gaussian state-space system:

$$y_t = Z\alpha_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, H) \quad (7)$$

$$\alpha_t = T\alpha_{t-1} + R\eta_t, \quad \eta_t \sim \mathcal{N}(0, Q) \quad (8)$$

where  $\alpha$  is the latent state vector (e.g., level and trend components),  $H$  is the measurement error covariance matrix, and  $Q$  is the innovation covariance matrix for the state shocks. These are learned from the data via a hierarchical prior specification, as described below.

#### Prior Elicitation with Maximum Entropy Transformations

To reflect informed but flexible beliefs about key smoothing and structural parameters, we employ maximum entropy-transformed priors (pz.maxent) for most hyperparameters. This allows us to specify constrained, information-efficient priors that respect known bounds or regularities (e.g., positivity, bounded damping).

#### Multivariate Covariance via LKJ Prior

We place an LKJ prior on the Cholesky decomposition of the state innovation covariance matrix  $Q$ , allowing for correlated innovations across state dimensions (e.g., co-movements between economic and climate variables). Specifically, we decompose  $Q = LL^T$ , where  $L$  is the lower triangular Cholesky factor, and the standard deviations and correlations are separately inferred. We set the concentration parameter  $\eta = 4$ , favoring weakly correlated structures but allowing for flexibility in learning strong correlations if supported by data. The standard deviation terms are assigned Gamma priors constrained to lie within plausible bounds.

#### Forecasting

**Posterior Inference and Inverse Transformation:** The model is then fit to data using Hamiltonian Monte Carlo (HMC) sampling via the nutpie backend (Langer and Developers, 2023) with jax acceleration (Bradbury *et al.*, 2018). The forecasts and trajectories generated by the model are returned in standardized space; hence, we apply an inverse exponential transformation composed with a StandardScaler inverse to convert model outputs back into the original data units.

#### 4.6. Country-level damage curves

Once we computed the probability distributions of event frequency, and the probability distribution of damages from points 4.2 and 4.3, the expected damages can be easily computed as a product of both:

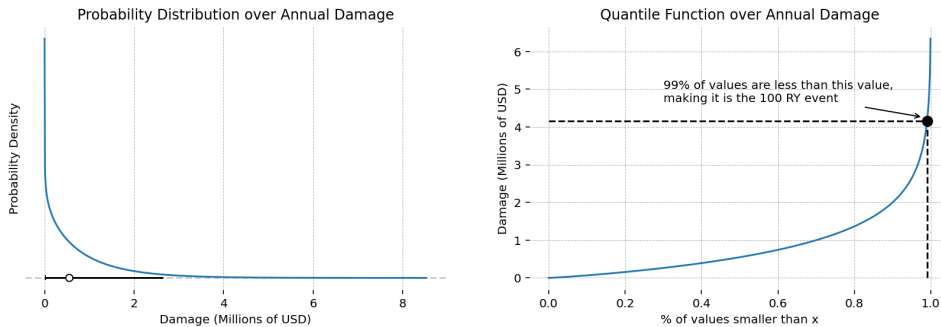
$$P(\text{expected\_damage})_{i,t} = P(\text{frequency})_{i,t} \times P(\text{damages})_{i,t} \quad (9)$$

One advantage of this approach, and the Bayesian computations method we use, is that it allows us to propagate and quantify uncertainty from the initial two models to the expected damages.

Now that we have the expected damages, we resort to the notion of **return years** (RY) to represent the results in a much more intuitive way. A 100-RY disaster is one the associated damage of which is expected to be equaled or exceeded once every one-hundred years. Our primary object of study is, thus, a damage curve expressed in return years. The left panel of Figure 3 shows an example distribution over damages for a given year. The x-axis shows possible damage values, while the height is the probability density at that damage. Based on this curve, we can create a quantile function, which maps probabilities (on the x-axis) to the damage value which is larger than that percentage of values. The 100-RY damage value is larger than 99% of all damage values produced by this distribution, and is marked.

Figure 3. Example Damage Curve for a Single Year and Disaster Type.

The left curve shows all possible damage outputs for a category of disaster (e.g. flood, earthquake, drought, etc.). The x-axis shows the possible dollar-denominated damages, while the y-axis shows the probability density associated with that damage value (higher = more probable). The black bar shows the largest interval containing 95% of all possible outcomes (called the 95% HDI). The right curve shows the quantile of the distribution on the left. The 100-RY disaster, associated with the 99% quantile, is marked.



Several features of Figure 3 are worth pointing out. The curve has a lot of mass around zero, followed by a long tail. Large disasters are, after all, relatively rare, with many years seeing none at all. As a result, the mean of the distribution



(marked with an open circle on the left plot) is not at all interesting from a policy perspective. Instead, we are interested in quantifying the density in the *tail*. Note that the 100-RY disaster, with a value of 4.1, is already a very rare event. But we can also contemplate even rarer events. In this example, a 500-RY event would have a value of 5.68. As the tail of the distribution becomes fatter, the difference between sizes of these rare events becomes larger.

Of primary interest to us is how climate change can potentially shift these curves, and how they vary according to geospatial patterns.

Climate change can shift these curves through two channels: disasters can become *more frequent* (increasing the probability of occurrence  $\mu$ ) or *more severe* (shifting the damage distribution conditional on occurrence). The resulting damage distribution is a zero-inflated mixture: with probability  $1 - \mu$  no disaster occurs and damage is zero; with probability  $\mu$  a disaster occurs and damage follows the severity distribution. Expected damages are simply  $\mathbb{E}[D] = \mu \cdot \mathbb{E}[D|I = 1]$ , the product of frequency and average severity. Both channels increase total expected damages, and the causal framework presented in Figure 2 justifies the statistical models we employ to estimate each component separately.

#### 4.7. Geolocated damage curves

Once we have the country-level expected damages, we disaggregate these estimates spatially by leveraging our previously estimated probability of disaster occurrence at each geolocated grid point  $g$ . This allows us to construct geolocated damage curves, capturing how expected damages vary across space within each country. The key idea is to assign a share of national expected damages to each location, proportional to its relative disaster probability. This results in a spatially disaggregated risk map with expected damage curves for each grid point.

$$P(\text{expected\_damage})_{i,t,g} = \frac{P(Y = 1|X)_{i,g,t}}{\sum_g P(Y = 1|X)_{i,g,t}} \times P(\text{expected\_damage})_{i,t} \quad (10)$$

This approach maintains internal consistency with the national-level expected damage figures. It distributes damages in proportion to the local probability of experiencing a disaster, as computed using our HSGP-based spatial event model. The denominator ensures the probabilities sum to one within each country-year, making the spatial distribution interpretable as a relative risk-weighted allocation. Importantly, this method inherits the full uncertainty structure of the Bayesian event and damage models, allowing us to construct local credible intervals around the expected damages at each location.

Once the expected damages are mapped at the local level, we proceed in a similar manner as before to construct geolocated damage curves. That is, for each location  $g$ , we generate the probability distribution over damage levels using the joint Bayesian posterior of both event probabilities and damage given an event.

This yields a distribution  $f_{D,g}(d)$  for each point  $g$  from which we derive return-year quantiles (e.g., 100-RY or 500-RY events) for local areas.

Geolocated damage curves add critical granularity to the policy relevance of our framework. Different locations within a country face widely varying exposure and vulnerability profiles. For instance, coastal regions may face frequent flooding, while inland agricultural zones may suffer from drought. Our framework accounts for these spatial patterns by conditioning on local predictors such as distance to rivers, elevation, and population exposure, alongside the HSGP structure that learns spatial dependencies directly from the data.

Finally, a similar causal framework used to interpret country-level changes under climate scenarios applies locally. Increases in  $P(Y = 1|X)_{i,g,t}$  and  $P(\text{events})_{i,t}$  derive in more frequent local disasters (e.g., more floods in coastal towns), while changes in  $P(\text{damages})_{i,t}$  imply higher severity conditional on a disaster (e.g., stronger typhoons hitting a specific region). Spatially resolved return-year curves provide the basis for identifying which areas are most vulnerable, how risks change under counterfactual scenarios, and where targeted adaptation investments may be most needed.

## 5. Data

### 5.1. Data sets summary

To process the data for the project, we built an open source automatized pipeline that downloads and processes the data. The pipeline can be accessed in the project's repository [link](#). We want to make this data pipeline publicly available as an exercise of transparency, to facilitate replication and because we are engaged promoters of open-source software.

The data sets used to build the Probabilistic GeoRisk model can be classified into three different categories:

#### 1. Country-level information

Following the approach of López *et al.* (2015), we use the World Bank World Development Indicators (The World Bank, 2024b), which contain a panel of yearly information for almost all countries in the world. From this data set, we extract each country's GDP per capita, population density, and total population.

To obtain each country's deviation from its trend precipitation, we use the GPCC world precipitation data set (Schneider, 2022), which contains geospatial information all around the world. The data set provides the monthly total precipitation on a regular grid with a spatial resolution of  $0.25^\circ \times 0.25^\circ$ ,  $0.5^\circ \times 0.5^\circ$ ,  $1.0^\circ \times 1.0^\circ$ , and  $2.5^\circ \times 2.5^\circ$  latitude by longitude. Using each country's shapefile, we transform this geospatial information into a country-level time series from which we calculate the trend and the deviation with respect to that trend.

We also use the EM-DAT data set (Centre for Research on the Epidemiology of Disasters, 2024) to obtain the total number of disasters per country and the economic damages they caused. As found by Botzen *et al.* (2019), other data sets like NatCatSERVICE and Sigma provide detailed information about disasters, especially in terms of disaster geolocation. Nevertheless, we decided to use the EM-DAT database given its public availability, and its status as the academic standard.

When working with EM-DAT disasters register, it is important to take into account possible reporting biases, as disasters with less severe impacts may not have been registered. Additionally, it is possible that as the reporting capacities of countries increase, the rate of registration of these minor events changes. To address this, we include some initial filters by considering only disasters after 1980 that affected more than 1,000 people according to the reported impact. We also limit the analysis to hydrometeorological disasters, which comprise storms, floods, and movements of wet masses. By applying these filters, the number of disaster observations is reduced from 26,664 to 5,692.

#### 2. Global-level climate time series

At the world time series level, we use NOAA's (National Oceanic and Atmospheric Administration) atmospheric CO<sub>2</sub> concentration data set (NOAA, 2024), which provides the worldwide levels of CO<sub>2</sub> atmospheric concentration in Gigatons. We also obtain the world's mean ocean temperature and its deviation from trend from the NOAA's Global Surface Temperature Dataset (Zhang *et al.*, 2019).

### 3. Geospatial data sets

In terms of geospatial information, we use the shapefiles from the World Bank Official Boundaries (The World Bank, 2024a), which contain the official maps of the territory and administrative divisions for each country. We also use the world river data set from HydroSHEDS (Lehner and Grill, 2013), which contains the maps of all the rivers around the world. Additionally we take advantage of the geospatial coordinates for disasters contained in the EM-DAT data set (Centre for Research on the Epidemiology of Disasters, 2024), which provide the exact geographical zone affected by each event.

In terms of geospatial information the EM-DAT data set has an important limitation, as most of the disasters are missing the coordinates data. Nevertheless, practically all of them have the exact name or names of the place where the disaster took place. To overcome this limitation, we developed an automatic query for an artificial intelligence model which transformed the location into exact coordinates. Details about this procedure are given in the current section.

The data sources used in our modeling pipeline are:

- **EM-DAT (CRED)**: Historical database of disaster occurrences and impacts worldwide.
- **NOAA Atmospheric CO<sub>2</sub>**: Records of atmospheric carbon dioxide levels over time.
- **NOAA Ocean Surface Temperature**: Historical measurements of ocean surface temperatures.
- **GPCC Precipitation**: Records of precipitation at various levels of geographic resolution.
- **HydroSHEDS (WWF)**: High-resolution geospatial data of river networks with flow data.
- **World Development Indicators (World Bank)**: Economic, social, and environmental indicators.
- **World Bank Shapefiles**: Maps of country and administrative borders.

### 5.2. Event frequency data

For the event frequency data, we transform the EM-DAT dataset to obtain a count of the total number of disasters per country per year. To do this, we assume that for countries with no recorded disasters in a given year, the count is zero. We then merge this dataset with the World Bank Development Indicators (The World Bank, 2024b), the CO<sub>2</sub> data from NOAA (2024), ocean temperature data from Zhang *et al.* (2019), and the global precipitation dataset from Schneider (2022).

As previously mentioned, the precipitation data is provided at a geospatial level. Therefore, we use the World Bank shapefiles to assign each point to its corresponding country. We then compute the average annual precipitation per country, along with its deviation from the long-term trend. For the ocean temperature data, we similarly compute deviations from the historical mean. Table 2 presents summary

statistics for the disaster event counts.

Table 1. Summary Statistics of Hydrological Disasters

| Statistic          | Value   |
|--------------------|---------|
| Count              | 5818.00 |
| Mean               | 0.79    |
| Standard Deviation | 1.83    |
| Minimum            | 0.00    |
| 25th Percentile    | 0.00    |
| Median             | 0.00    |
| 75th Percentile    | 1.00    |
| Maximum            | 28.00   |
| Kurtosis           | 54.00   |
| Skewness           | 5.90    |

*Note:* This table presents the summary statistics for hydrological disaster data, including count, mean, standard deviation, percentiles, and measures of kurtosis and skewness.

In Figure 4, we show the historical mean across all countries for the number of hydrometeorological events reported per year. A clear upward trend is observed. This may be partly due to reporting bias: as countries improve their institutional capacity, the number of recorded disasters tends to increase. However, it is important to note that we have already applied a filter to mitigate this bias by including only disasters that affected more than 1,000 people and occurred after 1980.

It is important to highlight that the GPCC precipitation dataset is only available up to 2020. To address this limitation, we estimate the model using data from 1980 to 2020. For predictions in more recent years, we use actual observed data for all covariates except precipitation, for which we rely on forecasts generated by a time series model.

Table 3 presents a general summary of the data used as covariates in the model.

### 5.3. Disaster damage data

Table 4 presents the summary for the damages data used for the damages model. One important limitation is that the number of disasters on the EM-DAT data set for which there is a reported damage value is considerably lower. As a consequence, the number of observations for the damage model is less than half of those in the event one: 2,300 vs. 5,692.

EM-DAT data set provides the damage values adjusted to 2011 USD, we adjust them to 2025 using the consumer price index time series from the U.S. Bureau of Statistics (U.S. Bureau of Labor Statistics, 2024).

Figure 4. Yearly mean across countries of the number of hydrometeorological disasters

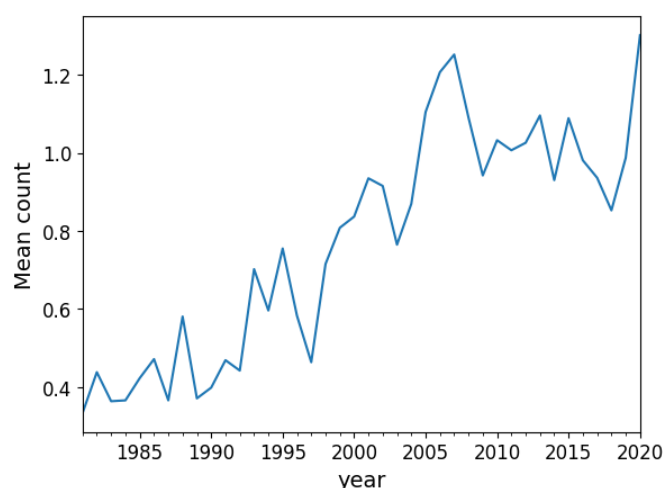


Table 2. Summary Statistics of event model covariates

| Statistic | Log of population density | Log of GDP per capita | Atmospheric CO2 | Population | Precipitation, deviation from trend |
|-----------|---------------------------|-----------------------|-----------------|------------|-------------------------------------|
| Count     | 5737.00                   | 5737.00               | 5737.00         | 5737.00    | 5737.00                             |
| Mean      | 3.81                      | 8.18                  | 374.92          | 41.23      | 0.34                                |
| Std Dev   | 1.33                      | 1.46                  | 21.31           | 140.71     | 171.03                              |
| Min       | 0.12                      | 5.10                  | 340.12          | 0.12       | -1477.12                            |
| 25%       | 2.94                      | 7.05                  | 356.54          | 3.69       | -69.37                              |
| Median    | 4.01                      | 8.09                  | 373.45          | 9.45       | -1.17                               |
| 75%       | 4.68                      | 9.20                  | 391.85          | 28.34      | 69.37                               |
| Max       | 7.15                      | 11.64                 | 414.21          | 1411.10    | 1576.26                             |
| Kurtosis  | -0.17                     | -0.86                 | -1.12           | 60.62      | 8.96                                |
| Skewness  | -0.40                     | 0.23                  | 0.17            | 7.53       | 0.09                                |

As seen in Figure 5, the damages data is characterized by extreme values reported in the biggest global economies like the United States, China and Japan. One clear example of this is the damages reported for hurricane Katrina in 2005, which have a 2025 adjusted value of 262 USD billions.

In contrast with the number of events registered, the damages of hydrometeorological disasters do not show a clear upward trend, as seen in Figure 6. Data reveals a peak of damages reported in 2005, a year where a set of highly destructive events were reported worldwide, including hurricane Katrina.

Table 5 presents the descriptive statistics for the features used in the damage model. In this case, to reduce the computing capacity required to estimate and sample the model, we standardize the features by subtracting the mean and dividing by the standard deviation.

Table 3. Summary Statistics of Total Damage Adjusted

| Statistic          | Total damage adjusted |
|--------------------|-----------------------|
| Count              | 2300.00               |
| Mean               | 1721.12               |
| Standard Deviation | 8233.07               |
| Minimum            | 0.00                  |
| 25th Percentile    | 25.32                 |
| Median             | 172.87                |
| 75th Percentile    | 848.19                |
| Maximum            | 261939.92             |
| Kurtosis           | 507.95                |
| Skewness           | 19.01                 |

*Note:* This table presents the summary statistics for Hydrometereological disasters total damage adjusted in 2025 USD millions.

Figure 5. Kernel density for yearly damages of hydrometeorological disasters

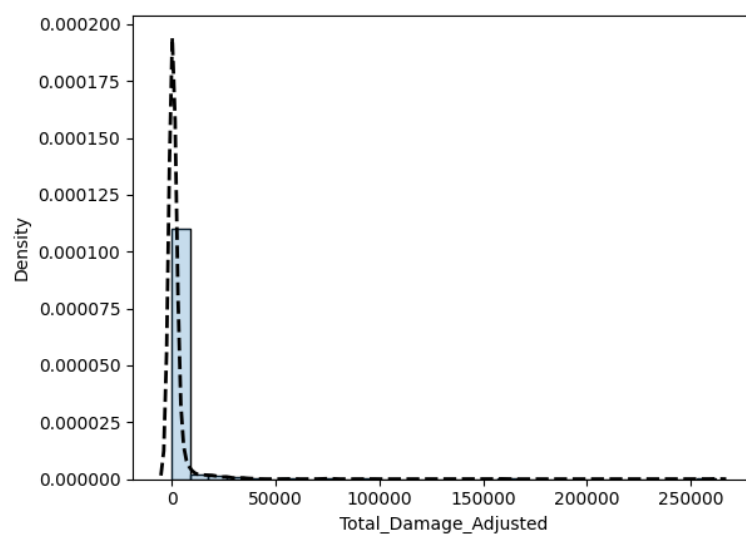
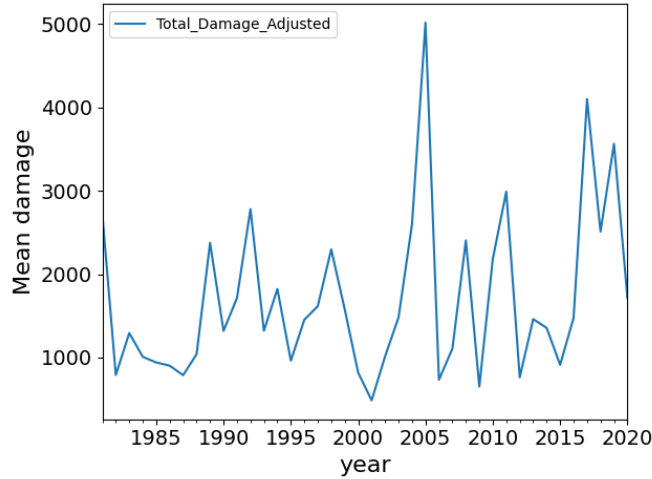


Table 4. Summary Statistics of damage model covariates

| Statistic | Log of population density | Log of GDP per capita | Log of GDP per capita, squared | Precipitation, deviation from trend | Atmospheric CO2 | Population |
|-----------|---------------------------|-----------------------|--------------------------------|-------------------------------------|-----------------|------------|
| Count     | 2300.00                   | 2300.00               | 2300.00                        | 2300.00                             | 2300.00         | 2300.00    |
| Mean      | 0.00                      | 0.00                  | 0.00                           | 0.00                                | 0.00            | 0.00       |
| Std Dev   | 1.00                      | 1.00                  | 1.00                           | 1.00                                | 1.00            | 1.00       |
| Min       | -3.28                     | -2.46                 | -1.99                          | -4.69                               | -1.99           | -0.69      |
| 25%       | -0.64                     | -0.68                 | -0.69                          | -0.37                               | -0.85           | -0.64      |
| Median    | 0.27                      | -0.14                 | -0.21                          | -0.11                               | 0.01            | -0.51      |
| 75%       | 0.79                      | 0.57                  | 0.50                           | 0.27                                | 0.86            | -0.09      |
| Max       | 2.08                      | 2.25                  | 2.48                           | 5.11                                | 1.74            | 2.15       |
| Kurtosis  | 0.59                      | -0.47                 | -0.30                          | 5.76                                | -1.05           | -0.04      |
| Skewness  | -0.90                     | 0.39                  | 0.69                           | 1.08                                | -0.06           | 1.34       |

*Note:* All features have been standardized.

Figure 6. Historical trend of yearly damages of hydrometeorological disasters



#### 5.4. Geospatial data

The backbone of the geospatial analysis is given by the World Bank country shapefiles (The World Bank, 2024a). Using the shapefiles, we can match geospatial information like latitude and longitude, with country-level information.

##### Use of Artificial Intelligence to complete locations

The use of EM-DAT disaster data at the geospatial level faces a significant challenge: out of 5,692 hydrometeorological disasters for which we have information, only 864 have associated geospatial coordinates. In contrast, 5,640 observations include data in the “Location” field, which typically contains the names of the places affected by the event in text format. It is important to note that this field often includes multiple locations, as a single disaster can impact several areas simultaneously. The key challenge, therefore, is to convert this location information from text format into geospatial coordinates.

To address this, we developed an automated query system that sends the location information to a large language model—in this case, OpenAI’s GPT-4 mini (OpenAI, 2024)—and requests the corresponding geographic coordinates. We then apply a validation step to ensure that the returned coordinates fall within the correct country boundaries. After processing, we are able to transform the textual location data into 2,876 individual coordinate points associated with the recorded events. Figure 7 compares in a map the geospatial representation of disaster locations before and after the query.

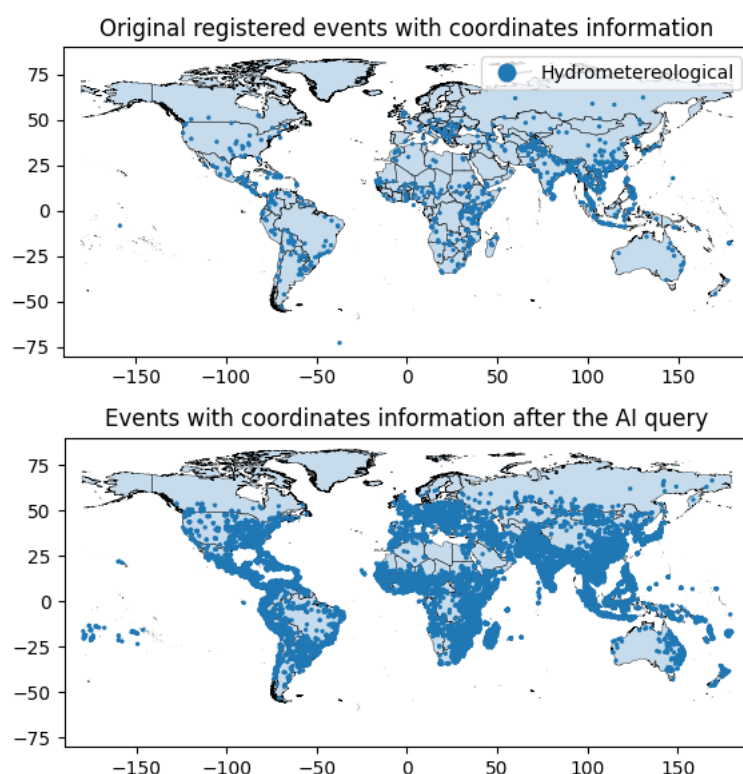
Figure 8 presents the map for Lao with the exact location and year of the hydrometeorological disasters after the query done with Artificial Intelligence.

##### River data set

For the rivers, we use a similar approach. From the HydroSHEDS database



Figure 7. Disasters with coordinates before and after the AI query



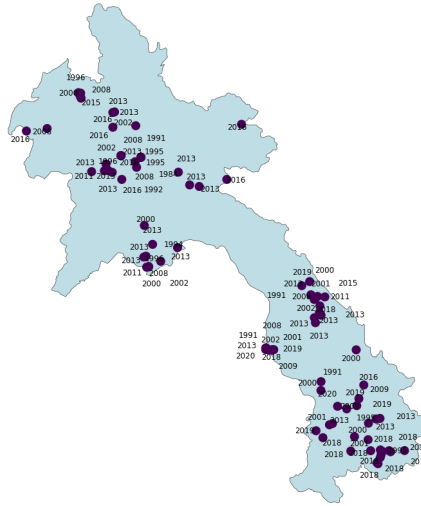
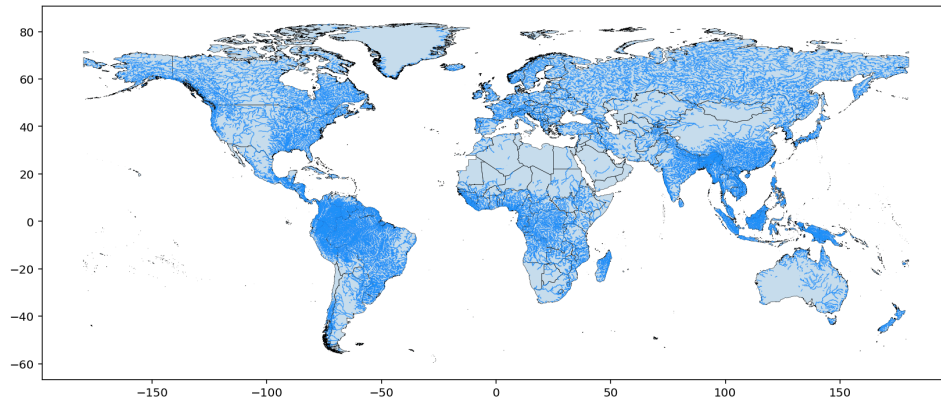
(Lehner and Grill, 2013), we obtain the coordinates for all the rivers around the world. Using one criterion, we filter the rivers to have those with the greater probability of causing floods: we include only the rivers with a long-term average discharge greater than  $100 \text{ m}^3$ . Which refers to rivers that have in the data set the ORD\_FLOW variable smaller than 5. Figure 9 presents the world mapping of the river data set used.

#### 5.4.1. Local data sets for geospatial modeling

In contrast to the first two models, which are estimated simultaneously using data from all countries with available observations, the geospatial component is estimated using data from a specific region only, and predictions are generated for a single target country. This approach is necessary due to the substantial computational resources required to estimate the model at a global scale. To illustrate the methodology, we present results for Laos.

To train the model for Laos, we use data from the following countries: Cambodia, Thailand, Laos and Vietnam.

Figure 8. Disaster geographical distribution by year for Lao

Figure 9. World rivers with a long-term average discharge greater than  $100 \text{ m}^3$ 

### Generation of fake non disaster data points

To estimate the geolocated probability of a disaster occurring at a specific point, we generate a set of synthetic non-disaster points and train a model to distinguish between real disaster locations and these synthetic counterparts. For each disaster point within a country, we randomly simulate 15 non-disaster points.

It is important to note that the disaster-to-non-disaster ratio does not affect the resulting probability values, as the outputs are normalized by dividing each by the sum of all point probabilities. However, through extensive testing, we found that a

1:15 ratio is particularly effective for generating high-quality samples for the HSGP model.

We then merge the disaster and synthetic non-disaster points into a single dataset. For each point, we calculate the distance to the nearest river and the nearest coastline. We also assign to each point the relevant country-level covariates for the year in which the event occurred and we standardize them to facilitate the estimation of the model. The full set of covariates used (all standardized by subtracting the mean and scaling by the standard deviation) is:

- Log of distance to river.
- Log of distance to coastline.
- Population.
- Accumulated atmospheric CO2 in Gigatons.
- Country's precipitation deviation from its Trend.
- Ocean temperature deviation from its Trend.
- Log of population density.
- Log of population density squared.
- Log of GDP per capita.

Figure 10 shows the map of disaster and synthetic non-disaster points for the region. Since we generate 15 random non-disaster points for every disaster, the point density is lower in countries with fewer recorded disasters. It is important to note that this does not affect the model's estimates related to event frequency, as that aspect is handled by the event component of the model, not in the geospatial component described here.

We provide the descriptive statistics of the dataset used to train the model for Laos and its neighbors.

Table 5. Summary Statistics of the geo-spatial model data set for Cambodia, Thailand, Laos and Vietnam

| Statistic | Is real disaster | Log distance to river | Log distance to coast | Population | Atmospheric CO2 | Precipitation, deviation from trend | Ocean Temperature, deviation from trend | Log of population density | Log of GDP per capita |
|-----------|------------------|-----------------------|-----------------------|------------|-----------------|-------------------------------------|-----------------------------------------|---------------------------|-----------------------|
| Count     | 30030            | 30030                 | 30030                 | 30030      | 30030           | 30030                               | 30030                                   | 30030                     | 30030                 |
| Mean      | 0.06             | 0.00                  | 0.00                  | 0.00       | -0.00           | -0.00                               | -0.00                                   | -0.00                     | -0.00                 |
| Std       | 0.23             | 1.00                  | 1.00                  | 1.00       | 1.00            | 1.00                                | 1.00                                    | 1.00                      | 1.00                  |
| Min       | 0.00             | -7.92                 | -8.83                 | -2.35      | -1.66           | -2.48                               | -2.35                                   | -3.44                     | -1.92                 |
| 25%       | 0.00             | -0.46                 | -0.49                 | -0.27      | -0.88           | -0.74                               | -0.71                                   | -0.42                     | -0.81                 |
| Med       | 0.00             | 0.22                  | 0.29                  | 0.21       | -0.07           | -0.01                               | -0.09                                   | -0.03                     | 0.02                  |
| 75%       | 0.00             | 0.69                  | 0.75                  | 0.61       | 0.80            | 0.65                                | 0.91                                    | 0.82                      | 0.82                  |
| Max       | 1.00             | 1.66                  | 1.30                  | 1.49       | 1.86            | 2.81                                | 1.95                                    | 1.28                      | 1.62                  |
| Kurt      | 13.13            | 3.35                  | 3.36                  | 0.44       | -1.13           | -0.00                               | -0.46                                   | 1.52                      | -1.11                 |
| Skew      | 3.89             | -1.46                 | -1.57                 | -1.05      | 0.20            | 0.40                                | -0.01                                   | -1.17                     | -0.17                 |

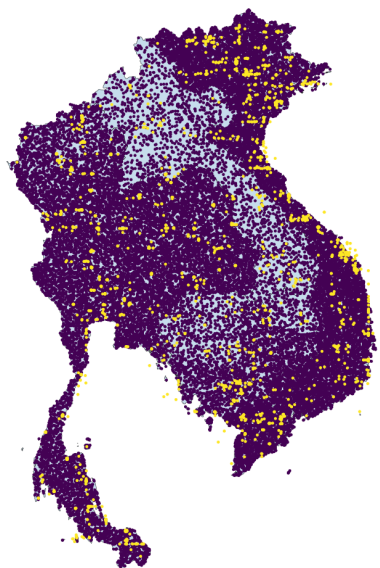
*Note:* All features, except for “is\_disaster” have been standardized.

### 5.5. Result grids

To visualize the model results and evaluate the probability of a disaster happening on every point of Laos, we project the country's map to a 400 x 400 grid. Then for each point of the grid we compute the distance to the closest river, and the distance

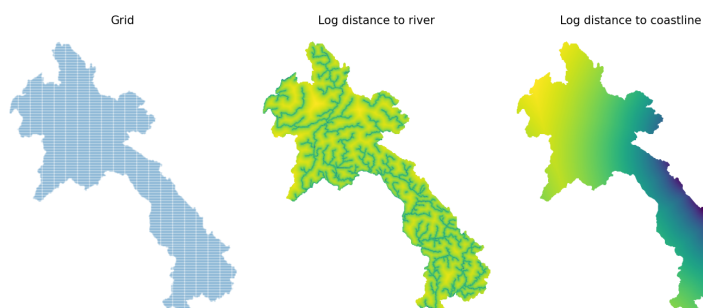
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Figure 10. Disaster and synthetic non-disaster points for Cambodia, Thailand, Laos and Vietnam. Yellow points correspond to reported disasters, purple to randomly generated non-disaster points.



to the nearest coastline point, then we merge that grid data set with the country level indicators of the year to evaluate. Figure 11 presents the grid and the plotting of the distance to rivers and coastline.

Figure 11. Laos grid with distance to rivers and coastlines



### 5.6. Climate and development time series

We model two types of time series: climate and development indicators in order to obtain future projections.

For the climate ones, we use:

- Total atmospheric CO<sub>2</sub> accumulation (in gigatons) (NOAA, 2024)
- Mean ocean temperature (Zhang *et al.*, 2019).
- Country level mean monthly precipitation (Schneider, 2022).

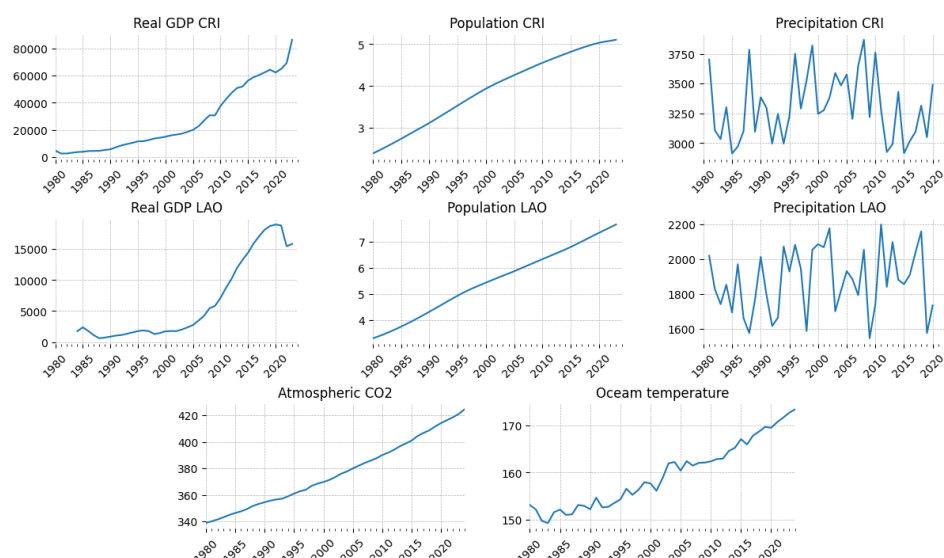
For the country development indicators, we extract from the World Development Indicators (The World Bank, 2024b):

- Real GDP in USD millions.
- Population in millions.

Figure 12 shows the temporal evolution of the variables included in the time series analysis. To deal with the trends and facilitate estimation, we standardize the data and compute the series first differences. Figure ?? in the annex presents the plots of this transformed data. The descriptive statistics for the transformed time series and their respective kernel density plots are presented in Table 11 respectively. It is necessary to comment that the Exponential Trend Smoothing (ETS) model used does not require the time series to be stationary.

Figure 12. Time series evolution through time.

Population is measured in millions, real GDP in millions of USD, CO<sub>2</sub> in Gigatons, ocean temperature deviation from its mean in Fahrenheit degrees, and precipitation deviation is the whole year monthly mean in millimeters.



## 6. Results

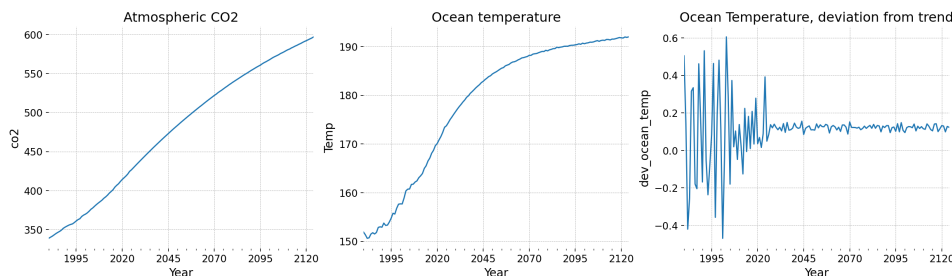
As a first step we present the results of the time series models. These results are extremely important as they allow us to project the covariates of the other models into the future, which is a requisite to produce predictions of the model results in the years to come.

### 6.1. Time series model results

The goal of the time series model is to generate predictions for the covariates used in the disaster frequency, disaster damages, and geospatial patterns models. This allows us to project the models into the future and evaluate the impact of different CO<sub>2</sub> scenarios on country-level damage curves. First, we present the results for the global-level time series of CO<sub>2</sub> and ocean temperature, along with the deviation of ocean temperature from its trend. We observe that both atmospheric CO<sub>2</sub> and ocean temperature are projected to continue their upward trends, though the growth rate of ocean temperature is expected to smooth over time.

Figure 13. Time series predictions for the global-level covariates.

CO<sub>2</sub> is measured in Gigatons, ocean temperature is measured in Fahrenheit degrees, and ocean temperature deviation from its mean is measured in standard deviation units.

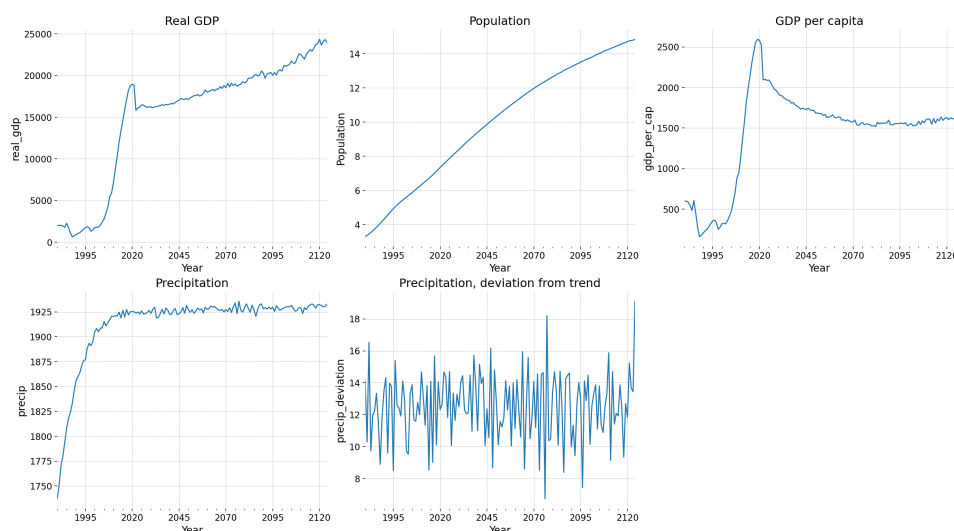


Regarding the country-level variables, Figure 14 presents the projections for Laos. We observe an upward trend in GDP and population. The projected population growth rate exceeds the GDP growth rate during a certain period, leading to a decline in GDP per capita. In the case of precipitation, the time series stabilizes after following the current trend for a few years.

### 6.2. Disaster frequency model

The sampling statistics of the frequency model are presented on tables 8 and 9. The sampling was implemented without divergences. The  $\hat{r}$  values for the results show a healthy sampling process. According to the Gelman–Rubin convergence diagnostic, the values of 1.00 mean that MCMC chains have converged to the same target posterior distribution. The only exception is the case of  $\beta_{GY}$ , which has

Figure 14. Time series predictions for Laos country level variables. Population is measured in millions, real GDP in millions of USD, and precipitation is the whole monthly mean in millimeters.



some  $\hat{r}$  values above 1.00. In general, the  $\text{ess\_tail}$ , which measures the effective sample size in the tails of the posterior distribution, reveal that we have reasonable sampling sizes, with the lowest one being 945.

Figure 15 presents the model results for Lao. Here the blue line provides the mean in-sample prediction, while the shaded region corresponds to the 95% Highest Density Interval (the darker the shade, the darker the probability for that value). We can see how the model succeeds to capture the general trend of the event's frequency, and how almost the observed number of events lie inside the 95% HDI values. In general, Lao presents a mean yearly number of events lower than one, with a trend to grow during the last decade, where the mean number of events increases up to one.

Figure 16 reveals a clear upward trend in the number of yearly events, with the mean value increasing up to 3. The 95% Highest Density Interval (HDI) extends up to 6 events. This means that by 2020, we could be 95% confident that the number of hydrometeorological disasters in Laos was fewer than 3. By 2060, however, this number could increase to 6. (It is important to highlight that, for the purposes of this study, we are only considering events reported to affect more than 1,000 people).

We can conclude then that Laos is exposed to an increase of the number of hydrometeorological events in the years to come according to our model.

Table 6. Summary Statistics 1 for frequency event model estimation

| Parameter                        | mean   | sd    | hdi_3% | hdi_97% | ess tail | r hat |
|----------------------------------|--------|-------|--------|---------|----------|-------|
| alpha                            | 10.923 | 1.575 | 8.076  | 13.839  | 6169.0   | 1.0   |
| alpha_X[ln_pop_density]          | 0.002  | 2.400 | -4.666 | 4.614   | 6163.0   | 1.0   |
| alpha_X[ln_gdp_pc]               | -0.047 | 2.429 | -4.824 | 4.463   | 6167.0   | 1.0   |
| alpha_X[precip_deviation]        | 0.002  | 2.351 | -4.368 | 4.521   | 6562.0   | 1.0   |
| alpha_X[co2]                     | -0.006 | 2.405 | -4.545 | 4.826   | 6275.0   | 1.0   |
| alpha_X[population]              | 0.043  | 2.522 | -4.605 | 4.951   | 6086.0   | 1.0   |
| alpha_X.loc[ln_pop_density]      | -0.019 | 1.007 | -1.927 | 1.847   | 5732.0   | 1.0   |
| alpha_X.loc[ln_gdp_pc]           | -0.003 | 1.005 | -1.897 | 1.903   | 5927.0   | 1.0   |
| alpha_X.loc[precip_deviation]    | 0.000  | 0.981 | -1.833 | 1.844   | 6378.0   | 1.0   |
| alpha_X.loc[co2]                 | -0.001 | 1.008 | -1.909 | 1.881   | 5888.0   | 1.0   |
| alpha_X.loc[population]          | 0.014  | 1.000 | -1.777 | 1.943   | 6242.0   | 1.0   |
| alpha_X.offset[ln_pop_density]   | 0.008  | 0.907 | -1.770 | 1.643   | 5903.0   | 1.0   |
| alpha_X.offset[ln_gdp_pc]        | -0.004 | 0.898 | -1.637 | 1.729   | 6051.0   | 1.0   |
| alpha_X.offset[precip_deviation] | -0.007 | 0.886 | -1.671 | 1.692   | 6037.0   | 1.0   |
| alpha_X.offset[co2]              | -0.006 | 0.885 | -1.650 | 1.661   | 5974.0   | 1.0   |
| alpha_X.offset[population]       | 0.010  | 0.929 | -1.729 | 1.739   | 5775.0   | 1.0   |
| alpha_X.scale[ln_pop_density]    | 1.996  | 1.416 | 0.037  | 4.517   | 4929.0   | 1.0   |
| alpha_X.scale[ln_gdp_pc]         | 2.001  | 1.433 | 0.051  | 4.544   | 5176.0   | 1.0   |
| alpha_X.scale[precip_deviation]  | 1.979  | 1.398 | 0.034  | 4.452   | 5291.0   | 1.0   |
| alpha_X.scale[co2]               | 1.989  | 1.410 | 0.051  | 4.476   | 4705.0   | 1.0   |
| alpha_X.scale[population]        | 2.002  | 1.418 | 0.033  | 4.465   | 4370.0   | 1.0   |

Note: Variables contained: alpha and alpha\_X.

Table 7. Summary Statistics 2 for frequency event model estimation

| Statistic                 | mean   | sd    | hdi_3% | hdi_97% | ess tail | r hat |
|---------------------------|--------|-------|--------|---------|----------|-------|
| beta[ln_pop_density]      | 0.338  | 0.108 | 0.140  | 0.544   | 4237.0   | 1.00  |
| beta[ln_gdp_pc]           | -0.651 | 0.088 | -0.813 | -0.478  | 3745.0   | 1.00  |
| beta[precip_deviation]    | 0.002  | 0.000 | 0.002  | 0.003   | 6027.0   | 1.00  |
| beta[co2]                 | 0.025  | 0.002 | 0.022  | 0.029   | 4213.0   | 1.00  |
| beta[population]          | 0.010  | 0.001 | 0.008  | 0.012   | 5081.0   | 1.00  |
| beta_GY[ln_pop_density]   | -0.548 | 0.306 | -1.119 | 0.010   | 1263.0   | 1.01  |
| beta_GY[ln_gdp_pc]        | -0.559 | 0.298 | -1.120 | -0.001  | 945.0    | 1.01  |
| beta_GY[precip_deviation] | -0.574 | 0.307 | -1.143 | -0.010  | 1017.0   | 1.03  |
| beta_GY[co2]              | -0.526 | 0.305 | -1.092 | 0.048   | 846.0    | 1.02  |
| beta_GY[population]       | -0.550 | 0.295 | -1.074 | 0.027   | 1468.0   | 1.01  |
| country_loc               | -3.935 | 0.644 | -5.203 | -2.784  | 4829.0   | 1.00  |
| country_scale             | 0.340  | 0.193 | 0.024  | 0.686   | 3321.0   | 1.00  |
| rho                       | 0.878  | 0.138 | 0.595  | 1.000   | 1132.0   | 1.01  |
| sigma_X[ln_pop_density]   | 1.004  | 0.993 | 0.000  | 2.865   | 3529.0   | 1.00  |
| sigma_X[ln_gdp_pc]        | 1.002  | 1.016 | 0.000  | 2.843   | 3382.0   | 1.00  |
| sigma_X[precip_deviation] | 1.006  | 1.010 | 0.000  | 2.868   | 3393.0   | 1.00  |
| sigma_X[co2]              | 0.996  | 0.984 | 0.000  | 2.777   | 3547.0   | 1.00  |
| sigma_X[population]       | 1.006  | 1.023 | 0.000  | 2.896   | 3314.0   | 1.00  |
| zero_prob                 | 0.998  | 0.002 | 0.994  | 1.000   | 3568.0   | 1.00  |



Figure 15. Event Frequency model results for Lao.

Black points correspond to the observed numbers, blue line to the in-sample mean prediction, and shaded region to the 95% HDI, where lighter shadow reflects lower probability.

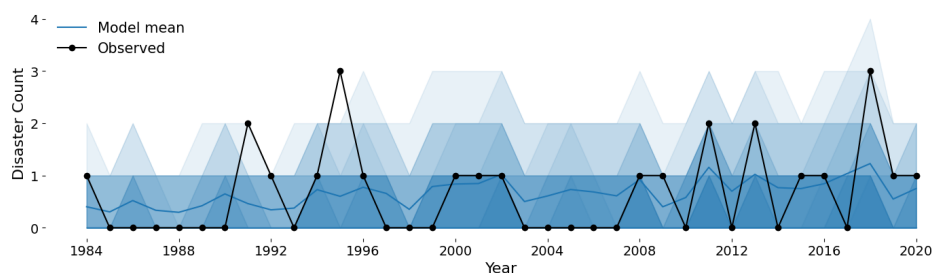
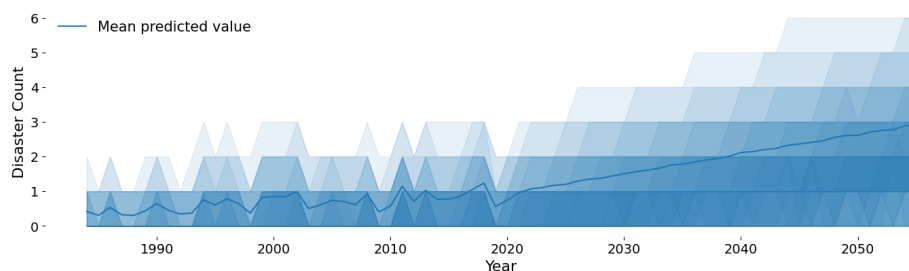


Figure 16. Event Frequency predictions for Lao.

Blue line to the mean prediction, and shaded region to the 95% HDI, where lighter shadow reflects lower probability.



### 6.3. Damages model

The sampling results for the damage model are presented on table 10, while the in sample model predictions are presented on Figure 17. The model was successfully sampled without divergences. From table 10 it is possible to observe that almost all variables have a  $\hat{r}$  of 1.00 which reveals a healthy posterior approximation. However three particular variables have  $\hat{r}$  values greater to 1.01:  $\alpha_X[\ln\_population\_density]$ ,  $\alpha_X[\ln\_gdp\_pc]$ , and  $\beta_{GX}[\ln\_population\_density]$ . This reveals the MCMC had some conversion troubles for those particular variables. However, all the variables remain under the 1.05 threshold.

The effective sample sizes ( $ess\_tail$ ) are generally high, indicating that the MCMC chains have mixed well and that the posterior distributions are well-characterized. The variables  $\alpha_X[\ln\_population\_density]$ ,  $\alpha_X[\ln\_gdp\_pc]$ , and  $\beta_{GX}[\ln\_population\_density]$  present the lowest  $ess\_tails$ , however the values are all equal or greater to 280, which still represent a reasonable size for posterior drawing.

Table 8. Summary Statistics of Model Parameters

| Parameter                      | mean   | sd    | hdi_3% | hdi_97% | ess tail | r hat |
|--------------------------------|--------|-------|--------|---------|----------|-------|
| alpha_X[ln_population_density] | -0.470 | 0.077 | -0.623 | -0.332  | 280.0    | 1.02  |
| alpha_X[ln_gdp_pc]             | -0.087 | 0.073 | -0.224 | 0.052   | 376.0    | 1.02  |
| alpha_X[precip_deviation]      | 0.621  | 0.086 | 0.464  | 0.781   | 1489.0   | 1.00  |
| alpha_X[co2]                   | 0.524  | 0.086 | 0.363  | 0.680   | 1366.0   | 1.00  |
| alpha_X[population]            | -0.590 | 0.075 | -0.730 | -0.447  | 544.0    | 1.01  |
| beta[ln_population_density]    | -1.047 | 0.494 | -1.898 | -0.057  | 3479.0   | 1.00  |
| beta[ln_gdp_pc]                | 0.413  | 0.185 | 0.062  | 0.760   | 3540.0   | 1.00  |
| beta[precip_deviation]         | 0.076  | 0.036 | 0.008  | 0.142   | 2549.0   | 1.00  |
| beta[co2]                      | 0.040  | 0.079 | -0.111 | 0.184   | 3116.0   | 1.00  |
| beta[population]               | -0.075 | 0.300 | -0.619 | 0.511   | 3245.0   | 1.00  |
| beta_GX[ln_population_density] | 0.308  | 0.010 | 0.288  | 0.328   | 688.0    | 1.03  |
| beta_GX[ln_gdp_pc]             | -0.007 | 3.109 | -5.797 | 5.695   | 2366.0   | 1.00  |
| beta_GX[precip_deviation]      | -0.007 | 3.052 | -5.386 | 6.063   | 2445.0   | 1.00  |
| beta_GX[co2]                   | 0.010  | 3.015 | -5.790 | 5.442   | 2686.0   | 1.00  |
| beta_GX[population]            | 0.028  | 2.993 | -5.584 | 5.513   | 3067.0   | 1.00  |
| beta_GY[ln_population_density] | 0.282  | 0.256 | -0.210 | 0.753   | 3415.0   | 1.00  |
| beta_GY[ln_gdp_pc]             | -0.056 | 0.231 | -0.503 | 0.375   | 3005.0   | 1.00  |
| beta_GY[precip_deviation]      | -0.169 | 0.230 | -0.635 | 0.235   | 2889.0   | 1.00  |
| beta_GY[co2]                   | -0.106 | 0.235 | -0.552 | 0.332   | 2778.0   | 1.00  |
| beta_GY[population]            | 0.047  | 0.242 | -0.397 | 0.520   | 2769.0   | 1.00  |
| country_loc                    | 2.616  | 0.301 | 2.062  | 3.185   | 2102.0   | 1.00  |
| country_scale                  | 0.145  | 0.118 | 0.000  | 0.367   | 1814.0   | 1.01  |
| p_zero                         | 0.002  | 0.001 | 0.000  | 0.004   | 2660.0   | 1.00  |
| rho                            | 0.959  | 0.055 | 0.861  | 1.000   | 1224.0   | 1.00  |
| sigma                          | 2.164  | 0.033 | 2.098  | 2.223   | 3026.0   | 1.00  |
| sigma_X[ln_population_density] | 0.125  | 0.002 | 0.121  | 0.128   | 2441.0   | 1.00  |
| sigma_X[ln_gdp_pc]             | 0.338  | 0.005 | 0.327  | 0.347   | 2674.0   | 1.00  |
| sigma_X[precip_deviation]      | 1.258  | 0.029 | 1.205  | 1.312   | 3285.0   | 1.00  |
| sigma_X[co2]                   | 1.259  | 0.029 | 1.208  | 1.314   | 3362.0   | 1.00  |
| sigma_X[population]            | 0.126  | 0.002 | 0.123  | 0.130   | 2733.0   | 1.00  |

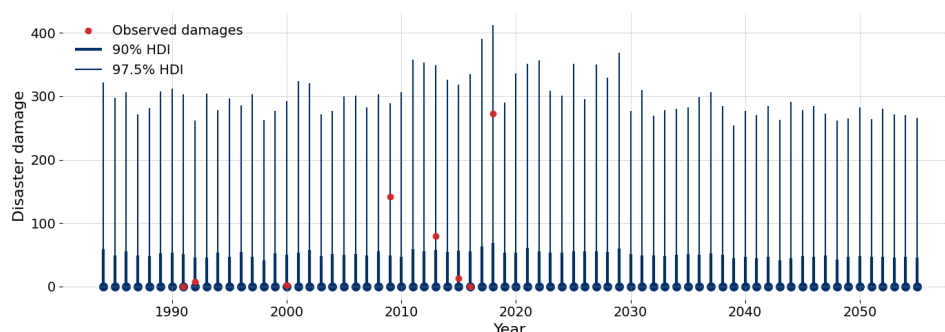
The damage model results and predictions for Lao are presented in Figure 17. This plot shows the in-sample damage predictions conditional on the occurrence of a disaster. We can observe how the country has experienced in the last decade three disasters that surpass in terms of damage all previous disasters. In that regard the 2018 flood was particularly destructive, with damages outside the 80% HDI. Interestingly, all the reported events are contained in the 95% HDI, with only two events outside the 80% HDI.

In terms of the projected values, the model does not project a growing trend in terms of the projected damages. Here it is important to take into consideration that the model has country-level hierarchy for different parameters, which means it infers different relations between covariates for countries.

It is necessary to underscore that the total exposure is measured by the interaction of damages and frequency, which we will explore in the next section.

Figure 17. Damage model results for Laos.

Red points correspond to the observed damages, light blue line corresponds to the 97.5% HDI, and wider blue line corresponds to 90% HDI. Damages are expressed in 2025 USD millions.



#### 6.4. Country level damages curve

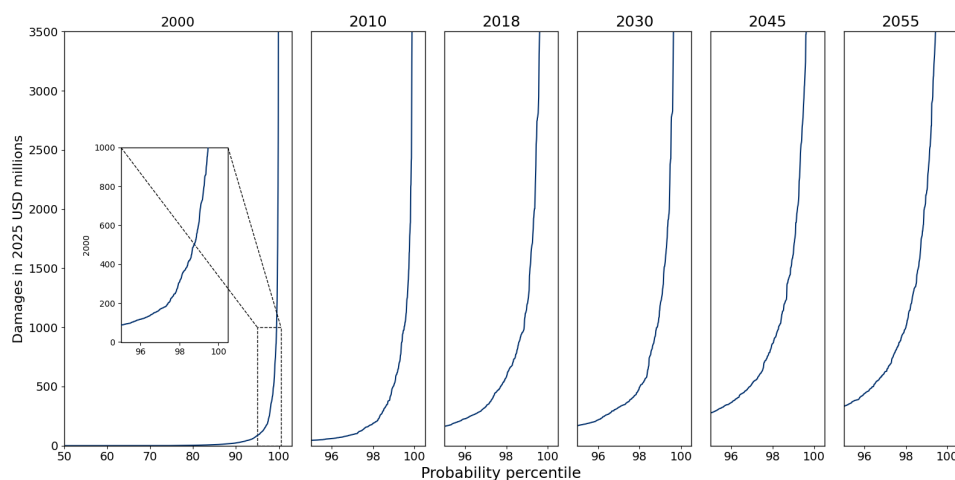
”To produce the damage curves, we combine the results of the damage and frequency models. By multiplying the estimated number of events by the expected damage, we obtain the total annual damage a country is projected to experience. The key advantage, however, is that we estimate the entire probability distribution rather than relying solely on mean values. This allows us to identify the specific damage level associated with every probability percentile. As discussed previously, hydrometeorological disasters typically exhibit heavy-tailed distributions, where sporadic, extreme events drive the majority of economic impacts. This is precisely where the capacity to model the full distribution becomes invaluable: it enables the direct evaluation of tail risks and an assessment of how the risk of extreme events evolves over time.

Figure 18 presents the damage curves for Laos. The vertical axis corresponds to damages in millions of USD, while the horizontal axis represents the probability associated with each damage level. As expected, most of the probability mass corresponds to low-damage events. However, as we consider less probable disasters, the magnitude of damages increases. The most relevant finding is that the model projects an upward shift in Laos’ damage curve, indicating that more impactful events will become more frequent, and more frequent events are projected to generate greater damages.

A more intuitive way of presenting the damage curves corresponds to the idea of **return years**. Table 12 and Figure 19 present the projected year returns for Lao. The projected return year values illustrate a clear escalation of climate-related disaster risks for Laos over time. While the frequency of smaller-scale events (e.g., 2-year and 1-year returns) remains negligible, the table shows a marked increase in damages for rarer, high-impact events. For instance, the 100-year return value rises from approximately USD 611 million in 2000 to over USD 2.1 billion by 2055.

Figure 18. Lao damage curves.

Damage curves are constructed by multiplying the probability distributions of disaster frequency by the expected damages. From the resulting joint probability distributions, we extract the tail risks to model the expected total damage a country faces during extreme events—specifically those in the higher percentiles. Each point on the curve reveals the damage associated with a specific probability percentile for a given year.



Similarly, the 50-year return value more than triples within the same period. This pattern underscores how the severity of extreme events is expected to intensify, placing growing economic and social pressure on the country.

Table 9. Projected Return Year Values by Year for Lao

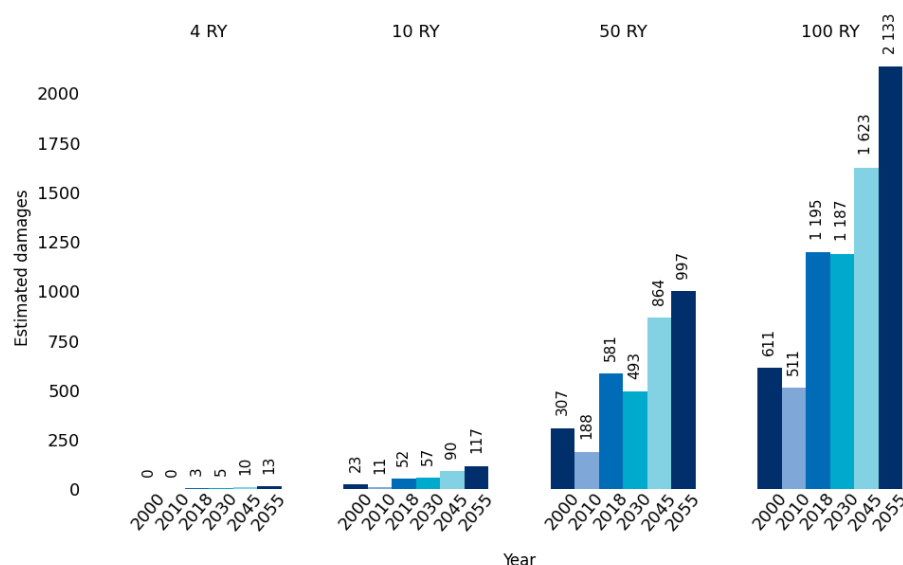
| Return Years | 2000   | 2010   | 2018    | 2030    | 2045    | 2055    |
|--------------|--------|--------|---------|---------|---------|---------|
| 100 RY       | 611.19 | 511.82 | 1195.83 | 1187.81 | 1623.36 | 2133.86 |
| 50 RY        | 307.86 | 188.15 | 581.92  | 493.46  | 864.37  | 997.77  |
| 10 RY        | 23.38  | 11.46  | 52.44   | 57.22   | 90.29   | 117.28  |
| 4 RY         | 0.72   | 0.00   | 3.47    | 5.39    | 10.58   | 13.53   |
| 2 RY         | 0.00   | 0.00   | 0.00    | 0.00    | 0.00    | 0.00    |
| 1 RY         | 0.00   | 0.00   | 0.00    | 0.00    | 0.00    | 0.00    |

*Note:* This table presents return year values for the specified years in 2025 USD millions.

Importantly, the results also suggest that moderate return periods (10-year and 4-year events) will increasingly contribute to the overall risk landscape. By 2055, a 10-year return event is projected to cause damages exceeding USD 117 million, compared to just USD 23 million in 2000. This shift means that not only will catastrophic events become more damaging, but recurrent disasters will also impose higher recurring costs.

Figure 19. Lao damage barplots.

Each set of bars represents the mean damage caused by a disaster that happens in average every X years, where X corresponds to the Return Years values. The Damage barplot illustrates how the mean damage for each Return Years category evolve through time



## 6.5. Geospatial patterns model

### 6.5.1. HSGP modeling

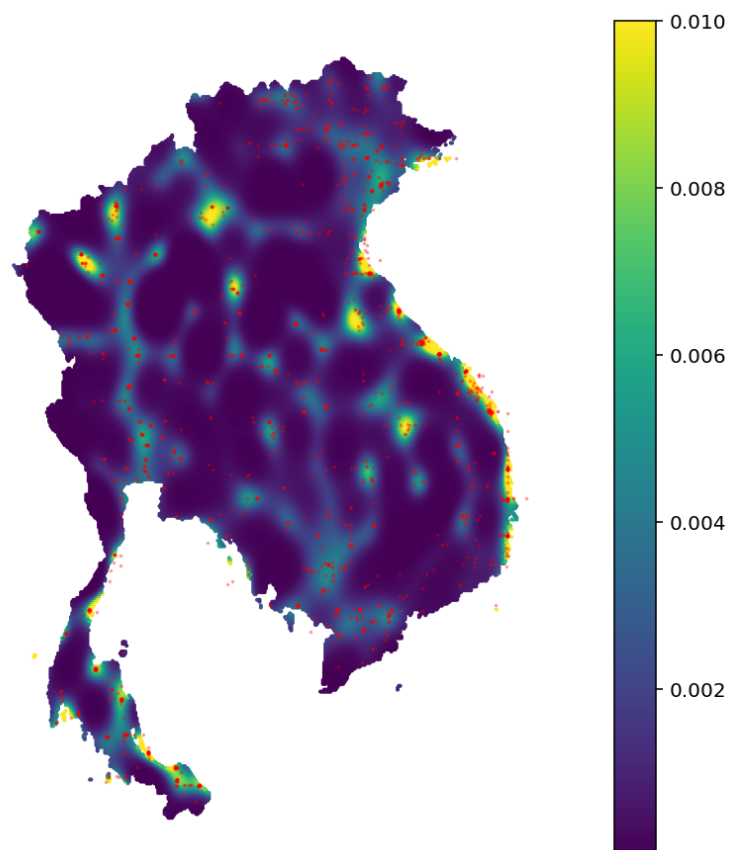
First, we implement a simplified geospatial model that relies solely on an HSGP component to estimate the probability of a disaster occurring at any given point within a country. This simplified HSGP version uses only latitude and longitude as independent variables, making it time-independent.

The results, shown in Figure 20, cover Cambodia, Thailand, Laos, and Vietnam. It is striking to see how the model can infer the outlines of major rivers purely from disaster occurrence patterns. We also observe that certain coastal regions exhibit greater exposure to disasters. This reinforces the idea that the HSGP component of the model represents a strong tool to account for omitted variables, as it is able to capture patterns of geographical patterns not explicitly included in the model. These results also confirm the notion that distance to rivers and coastlines represent a clear risk factor in the region.

HSGP parameter values for Cambodia, Thailand, Laos and Vietnam can be found in Annex 2.

Figure 21 presents the results specifically for Lao. Based only on the coordinates of previous disasters, the model is able to identify a risk zone in the center-left area of the country, precisely where a big number of rivers merge. It also identifies a risk

Figure 20. HSGP event probability predictions for Cambodia, Thailand, Laos and Vietnam. The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of one disaster. Red points correspond to observed disasters.



area closer to the capital Vientiane, and in two more points in the lower right area of the country.

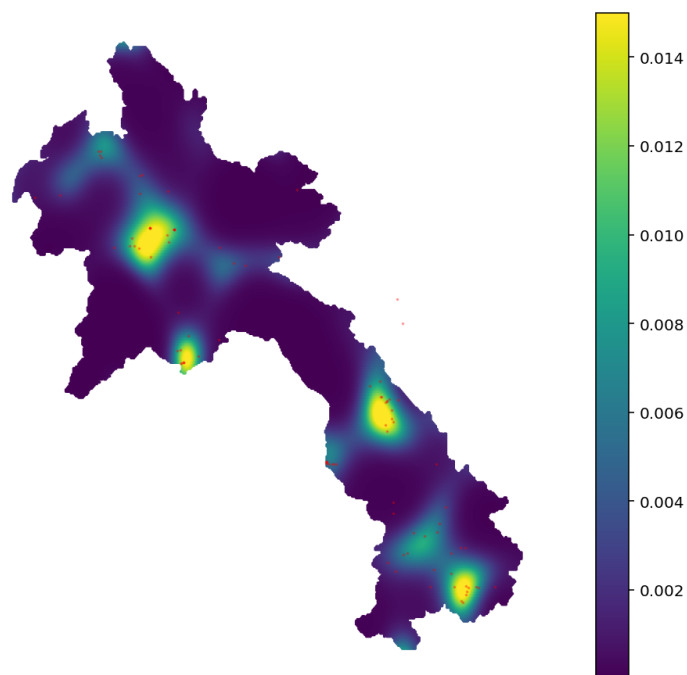
#### 6.5.2. Full model

Now, we present the estimated results for the full model, starting with Laos and its neighboring countries. It is important to highlight that we choose Lao and its neighboring countries as an example, nevertheless, the model can generate estimations and results for almost any country or region in the world for which the data is available.

For an easier understanding, we present the results in maps, however, the model produces specific probabilities for each point of the country, which can be used for

Figure 21. HSGP event probability predictions for Lao.

The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of one disaster. Red points correspond to observed disasters.



specific computations. We will present an example of this later.

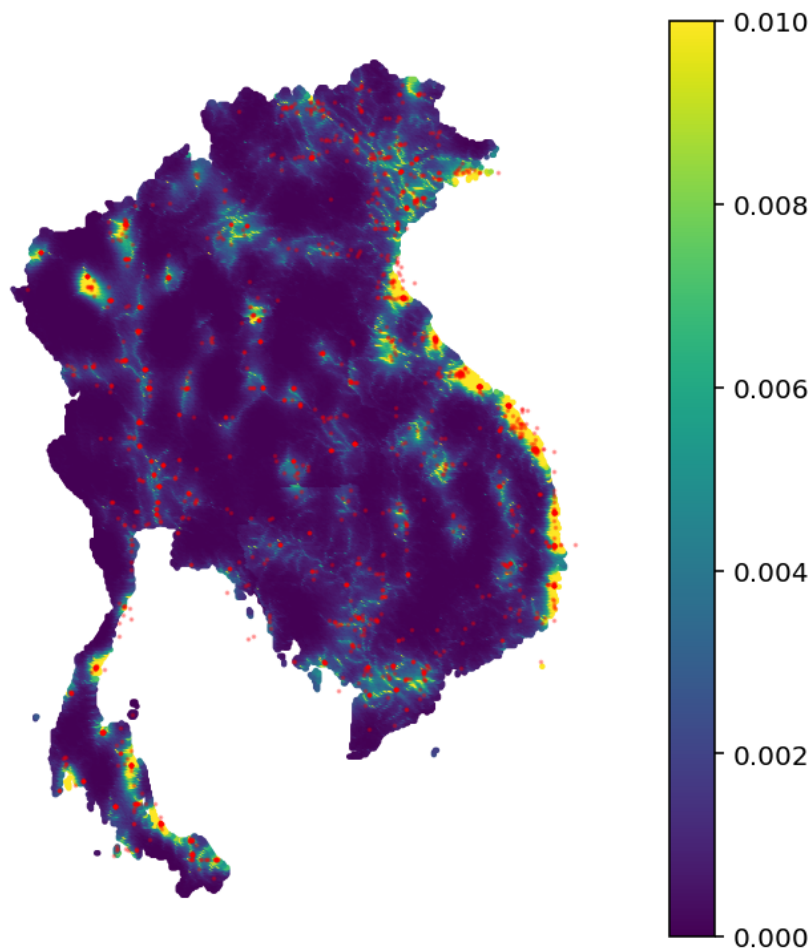
The main parameter values of the Full Model estimation can be found in Annex 2. Figure 22 shows the results for Cambodia, Thailand, Laos, and Vietnam. It is possible to observe that the distance to rivers is a relevant determinant of exposure to hydrometeorological events. However, it is not just the distance to any river that matters—the model combines the geospatial patterns identified by the HSGP with river distance information to determine high-risk areas. Consequently, some areas have a relatively low probability of events despite being close to rivers.

The same is true for distance to the coastline. The model is able to determine which coastal regions pose the greatest risk, as can be seen in the eastern part of the region. As mentioned earlier, this is due to the capacity of the HSGP component to identify and project geospatial patterns, which helps account for missing covariates.

We now zoom in to examine in more detail the specific results for Laos. Figure 23 presents the results for the country. Once again, we can see how the model is able to identify which rivers represent greater risks according to the geospatial patterns of past disasters.

Figure 24 presents the decomposition of the effects for Lao. It is important to

Figure 22. Full model event probability predictions for Cambodia, Thailand, Laos and Vietnam. The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of at least one disaster. Red points correspond to observed disasters.



consider that besides the effects plotted, the event probability risk is also affected by the country effect, the climate covariates and the country-level covariates, which do not vary at the geospatial level. The decomposition of the effects allows to identify that the HSGP component dominates the projected probability values. It also shows that despite having a lower weight, the distance to coastline has some positive impact, as can be seen in the lower right segment.



Figure 23. Full model event probability predictions for Lao.  
The plot value measures the yearly probability that a disaster will happen on each specific point conditional on the occurrence of at least one disaster. Red points correspond to observed disasters.

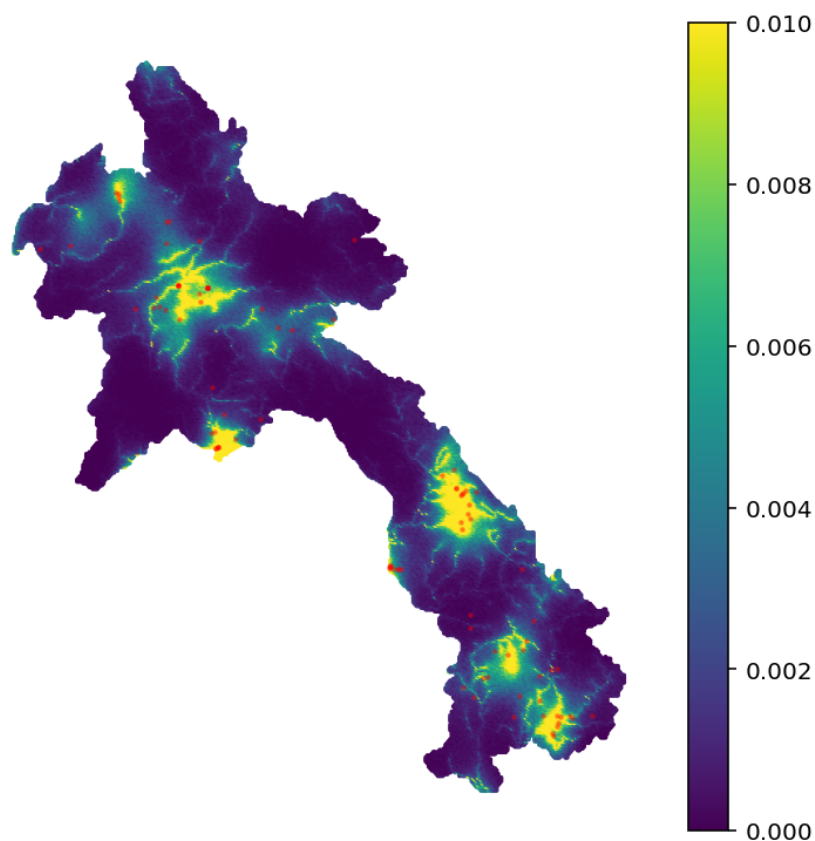
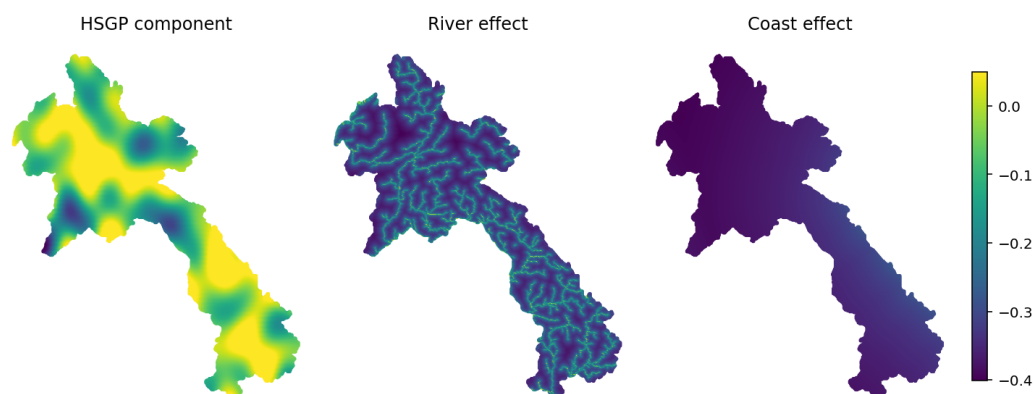
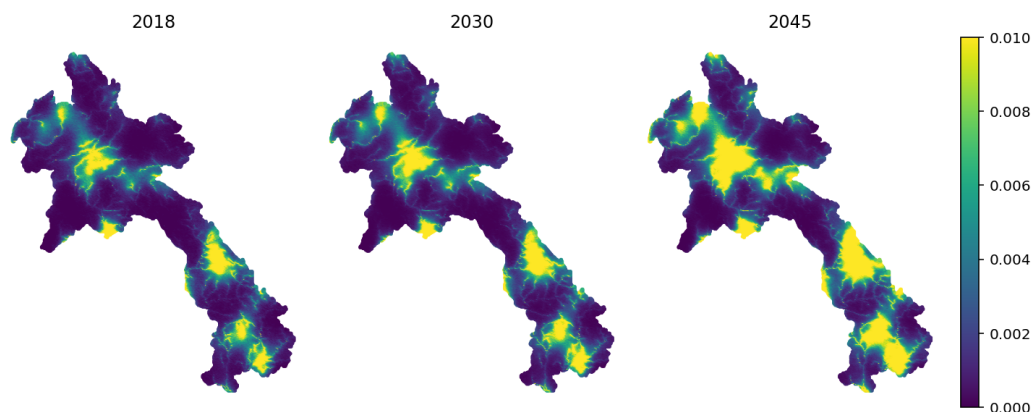


Figure 24. Decomposition of the event probability predictions for Lao.  
The decomposition is given in log-odds units



The static geospatial perspective is highly valuable in itself; however, given the normalization process applied to obtain the disaster probability, the value scale will remain relatively stable over time. We are interested in understanding how exposure to climate change risks evolves and how it is affected by projected climate trends. To gain a clearer perspective of how geospatial event risk evolves through time, we multiply the geospatial probability by the probability distribution of the country's event frequency; this provides the expected number of events for each point on the map. We then project these values to understand how risk evolves as atmospheric CO<sub>2</sub> increases. Figure 25 shows how local exposure increases over time for Laos.

Figure 25. Expected number of disasters for each point of Lao through time



### 6.6. *Geolocated return year curves*

One of the main contributions of this paper is to provide local-level estimates of event probabilities and the expected damages caused by climate change-related disasters. This approach is not intended to replace on-site hydrological studies, but rather to offer a low-cost and efficient tool that can provide initial approximations of risk.

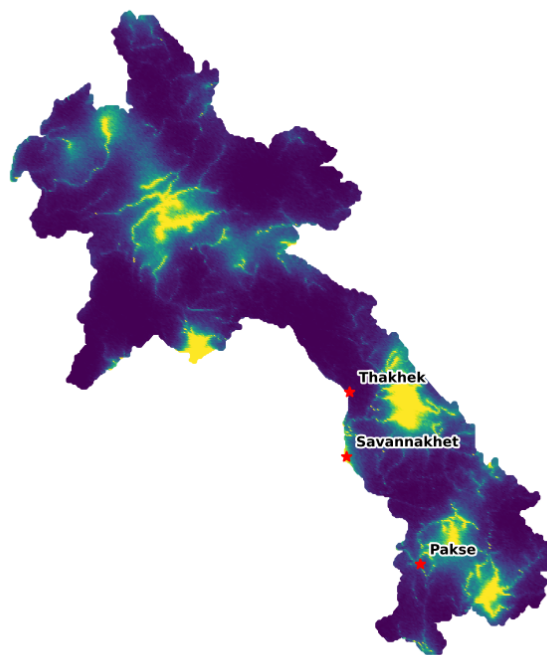
The model makes it possible to compute these values for any point within the analyzed country. We use the local probability to scale down the national expected damage, and produce local level return year curves. These curves do not consider local characteristics of infrastructure, or physical capital, but are local projections of the national damages using the local event probability (which does capture some of these patterns via the HSGP component).

These local projections allow us to quantify the climate change risk exposure at a granular level, and produce approximations of the impacts that particular communities can suffer due to climate change related disasters. To illustrate this, we focus on three specific locations in Lao PDR:

- Pakse.
- Savannakhet.
- Thakhek.

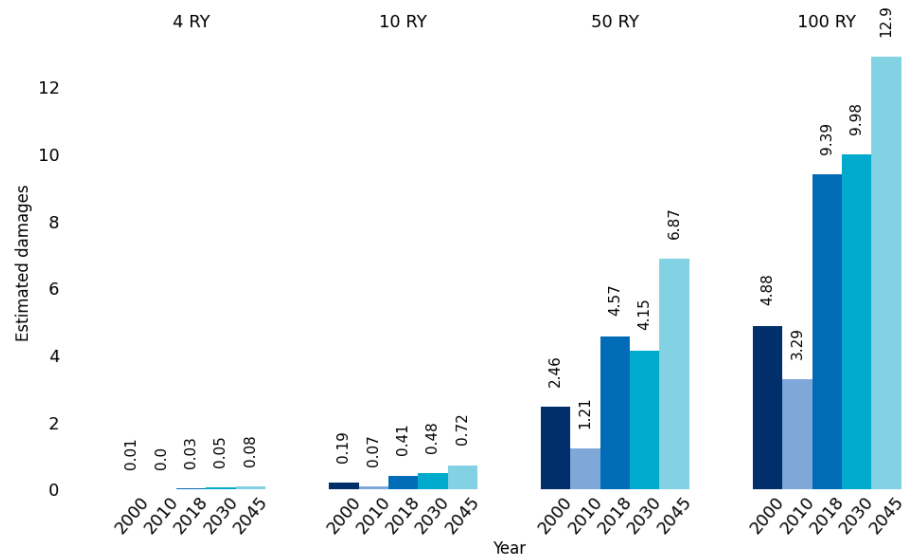
These locations are shown in Figure 26.

Figure 26. Selected locations in Lao PDR.



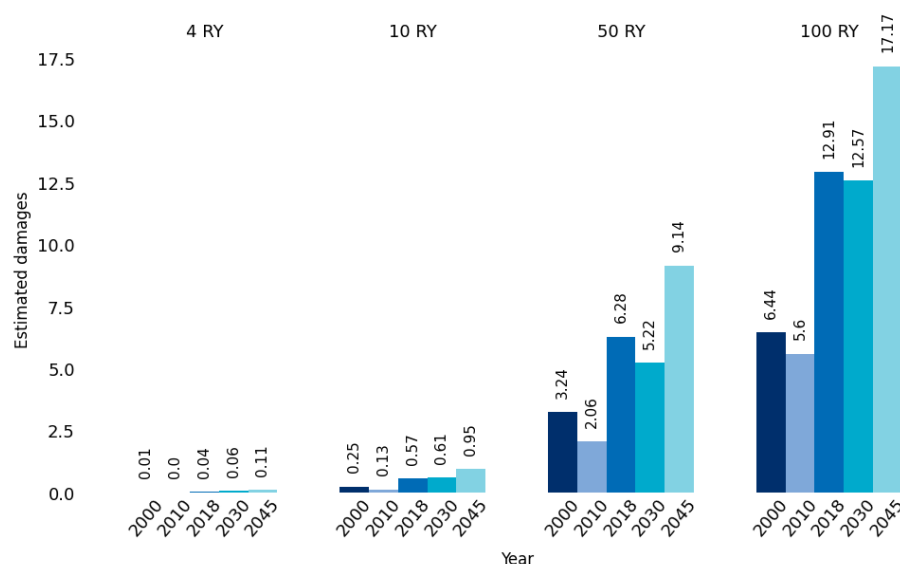
To exemplify the geolocated return year curves, we present in Figures 27 and 28 the return year curves for Pakse and Savannakhet respectively. Consistent with what has been observed in the previous results, the return year curves reveal a growing pattern in terms of damage exposures as more severe events become more frequent, and more frequent events become more severe.

Figure 27. Geolocated projected return year curves for Pakse.  
Estimated damages are measured in millions of 2025 USD.



For Pakse (Figure 27), the 100-year return damages are projected to grow from approximately USD 4.9 million in 2000 to USD 12.9 million by 2045. Even moderate return periods show a marked increase: the 50-year return value rises from USD 2.5 million to USD 6.9 million, while the 10-year return value grows from USD 0.2 million to USD 0.7 million over the same period.

Figure 28. Geolocated projected return year curves for Savannakhet. Estimated damages are measured in millions of 2025 USD.



Savannakhet (Figure 28), which presents the highest expected event frequency among the three locations, exhibits correspondingly higher projected damages. The 100-year return value reaches USD 17.2 million by 2045, compared to USD 6.4 million in 2000. The 50-year return value increases from USD 3.2 million to USD 9.1 million. These local projections underscore how the combination of high geospatial event probability and the projected increase in disaster frequency and severity translates into a substantial escalation of climate-related risk for specific communities in Lao PDR.

## 7. Conclusions

This study introduces the Probabilistic Geospatial Risk (PGR) model, an end-to-end framework designed to quantify and project the frequency, economic damage, and spatial distribution of climate change-related disasters. By integrating Bayesian generative modeling, state-of-the-art spatial analysis, and hierarchical time-series forecasting, the PGR model transcends traditional national-level risk assessments. It provides granular, point-level estimates of disaster risk, enabling policymakers, researchers, and practitioners to identify high-risk areas and prioritize interventions.

The model's ability to capture non-linear relationships and spatial autocorrelation through techniques like the Hilbert Space Gaussian Process (HSGP) and Intrinsic Conditional Autoregressive (ICAR) models provides valuable insights, while its use of generally available satellite information allows predictions even in data-scarce

regions.

### **From Global Trends to Local Action**

The PGR model’s outputs are highly actionable for climate adaptation and disaster risk reduction. By projecting return-year damage curves and geolocated risk maps, the model bridges the gap between global climate trends and local vulnerability. For example:

Laos was used as a case study to demonstrate how the model can identify regions with escalating exposure to hydrometeorological disasters. The results reveal that the country faces increasing frequencies of extreme events, with projected damages rising over the coming decades.

The geolocated return-year curves illustrate how communities can use these tools to assess their unique risks and plan targeted adaptations, such as infrastructure upgrades or early warning systems.

The model’s ability to disaggregate national-level risks into local probabilities empowers governments to allocate resources more effectively, focusing on areas where climate impacts are projected to be most severe.

A core innovation of the PGR model is its emphasis on uncertainty quantification. Traditional risk assessments often focus on point estimates (e.g., mean or mode), which are insufficient for policy planning. In contrast, the PGR model leverages posterior predictive distributions to simulate the full range of possible outcomes, including tail risks. This approach is critical for understanding the potential for catastrophic events, which, while rare, can have outsized economic and humanitarian consequences.

The use of Markov Chain Monte Carlo (MCMC) sampling and Bayesian hierarchical modeling ensures that uncertainty is propagated through all stages of the analysis, from frequency and damage modeling to spatial risk mapping. This transparency allows decision-makers to evaluate not just the most likely scenarios but also the worst-case possibilities, fostering more resilient planning.

The study’s projections underscore the urgent need for climate action. Under business-as-usual scenarios, Laos is expected to experience:

- Higher frequencies of hydrometeorological disasters, with return periods for extreme events shortening dramatically (e.g., 100-year events becoming 50-year events by mid-century).
- Increased severity of damages, as rising ocean temperatures and CO<sub>2</sub> concentrations amplify the intensity of storms, floods, and droughts.
- Spatial shifts in vulnerability, with regions close to rivers facing disproportionate risks due to geographic exposure.

These findings align with the broader scientific consensus that climate change will exacerbate disaster risks, but the PGR model provides the granularity needed to translate global trends into local action.

## Acknowledgments

## Appendix A.

Table 10. Descriptive Statistics of time series

| Statistic | Real GDP<br>CRI | Population<br>CRI | Precipitation<br>CRI | Real GDP<br>LAO | Population<br>LAO | Precipitation<br>LAO | Atmospheric<br>CO2 | Ocean<br>temperature |
|-----------|-----------------|-------------------|----------------------|-----------------|-------------------|----------------------|--------------------|----------------------|
| Count     | 44.00           | 44.00             | 40.00                | 40.00           | 44.00             | 40.00                | 45.00              | 45.00                |
| Mean      | -0.00           | -0.00             | 0.00                 | -0.00           | 0.00              | 0.00                 | -0.00              | 0.00                 |
| Std Dev   | 1.01            | 1.01              | 1.01                 | 1.01            | 1.01              | 1.01                 | 1.01               | 1.01                 |
| Min       | -1.77           | -2.03             | -1.50                | -1.64           | -1.95             | -1.92                | -1.57              | -1.53                |
| 25%       | -0.85           | -0.79             | -0.81                | -0.70           | -0.77             | -0.71                | -0.83              | -0.94                |
| Median    | -0.02           | 0.24              | -0.07                | -0.39           | 0.17              | 0.02                 | -0.10              | -0.10                |
| 75%       | 1.02            | 0.88              | 0.72                 | 1.11            | 0.83              | 0.93                 | 0.82               | 0.73                 |
| Max       | 1.58            | 1.26              | 1.94                 | 1.50            | 1.47              | 1.68                 | 1.85               | 1.95                 |
| Kurtosis  | -1.23           | -0.99             | -0.97                | -1.43           | -0.99             | -0.97                | -1.15              | -1.11                |
| Skewness  | -0.15           | -0.52             | 0.32                 | 0.27            | -0.40             | -0.21                | 0.20               | 0.30                 |

*Note:* This table presents descriptive statistics for climate time series included in the model. The series have been standardized, and then their first difference was computed.

## Appendix B.

Table 11. Summary Statistics of Model Parameters for HSGP estimation in Cambodia, Thailand, Laos and Vietnam

| Parameter    | mean  | sd   | ess bulk | ess tail | r hat |
|--------------|-------|------|----------|----------|-------|
| ell_log__[0] | -0.54 | 0.10 | 2027.00  | 2569.00  | 1.00  |
| ell_log__[1] | -0.62 | 0.08 | 2375.00  | 3180.00  | 1.00  |
| eta          | 4.99  | 0.36 | 1376.00  | 2432.00  | 1.00  |
| eta_log__    | 1.60  | 0.07 | 1376.00  | 2432.00  | 1.00  |

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Table 12. Summary Statistics of Full Model Parameters for Cambodia, Thailand, Laos and Vietnam

| Parameter                                     | mean  | sd   | ess bulk | ess tail | r hat |
|-----------------------------------------------|-------|------|----------|----------|-------|
| ell_log__[0]                                  | -1.37 | 0.25 | 3987.00  | 4003.00  | 1.00  |
| ell_log__[1]                                  | -1.35 | 0.18 | 4197.00  | 4835.00  | 1.00  |
| eta                                           | 3.12  | 0.36 | 2088.00  | 3751.00  | 1.00  |
| eta_log__                                     | 1.13  | 0.11 | 2088.00  | 3751.00  | 1.00  |
| beta[log_distance_to_river__standardized]     | -0.44 | 0.03 | 11561.00 | 5363.00  | 1.00  |
| beta[log_distance_to_coastline__standardized] | -0.17 | 0.06 | 7776.00  | 5073.00  | 1.00  |
| beta[Population__standardized]                | 0.10  | 0.09 | 14518.00 | 6600.00  | 1.00  |
| beta[co2__standardized]                       | 0.25  | 0.06 | 13074.00 | 6135.00  | 1.00  |
| beta[precip_deviation__standardized]          | 0.33  | 0.03 | 8833.00  | 5334.00  | 1.00  |
| beta[dev_ocean_temp__standardized]            | 0.02  | 0.03 | 8344.00  | 4750.00  | 1.00  |
| beta[log_population_density__standardized]    | 0.32  | 0.09 | 14313.00 | 6382.00  | 1.00  |
| beta[log_gdp_per_cap__standardized]           | 0.12  | 0.08 | 15245.00 | 6216.00  | 1.00  |
| country_effect[KHM]                           | -1.75 | 0.41 | 5900.00  | 5188.00  | 1.00  |
| country_effect[LAO]                           | -1.73 | 0.43 | 7930.00  | 6514.00  | 1.00  |
| country_effect[THA]                           | -4.52 | 0.26 | 6296.00  | 5787.00  | 1.00  |
| country_effect[VNM]                           | -3.87 | 0.30 | 5890.00  | 5449.00  | 1.00  |

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